P003.S01 — Failure of the primitive path integral at $r = 0$ for the Coulomb potential

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Solver: Claude Opus 5.5; auditor: Gemini; logged by `/log` v1.6

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Release gates: 3 of 14 failed. Stamped here by `/log` per `skills/log/SKILL.md` step 2, which requires a failed gate to be recorded in `meta.yaml` and in a banner at the top of this document rather than quietly fixed. The user closed the loop with these failures on record. None of the three touches a derivation step or a number.

`no_unflagged_recall` — fail. Two [R] tags are not in the Risk register (§7): the proton radius in §4.1 item 6, and "$R_{nl}(0) = 0$ for $l \geq 1$" in the coverage note of check C4. Neither is load-bearing. They are not added here, because step 3 of the log skill forbids editing the final document.

`code_audited` — fail. The audit's pass 6 is one line ("Code was audited by reading; execution not performed in this session"), and it reviews no check for circularity, vacuity or laundered assumptions. Check C6, added in round 2, was never seen by the auditor.

`latex_clean` — fail. This document compiles with no undefined references, no undefined citations, no missing characters and no `\emph`, but two overfull boxes exceed 10 pt: 39.1 and 34.8 pt. They are the display equations (A3) and (B10), which are wider than the text block. Both are typographic and are left as they are, for the same step-3 reason. Five overfull rows in the provenance table, which `/log` authors, were cleared by widening its columns.

The other eleven gates pass.

Abstract

The time-sliced (primitive) trace $Z_N(\beta) = \mathrm{Tr}(e^{-\Delta\tau K}e^{-\Delta\tau V})^N$ for hydrogen, $K = -\frac{1}{2}\nabla^2$, $V = -1/r$, is written as a $3N$-dimensional bead integral with every prefactor, and is shown to be $+\infty$ for every $N \geq 1$ and every $\Delta\tau > 0$. Two regions of configuration space are responsible. One is any single bead approaching the nucleus, where $e^{\Delta\tau/r}$ is not integrable. This divergence is specific to time slicing; the Gaussian kinetic factors cannot compensate it, and with a cutoff they do not change its leading growth. The other is the whole chain moving off to infinity, a divergence the exact trace shares. In the continuum, the exact kernel is finite and continuous everywhere, the nucleus included, because $-1/r$ is in the Kato class, while $\mathrm{Tr}\, e^{-\beta H}$ is infinite for the large-volume reason alone. The imaginary-time Trotter product formula requires both summands to be bounded below, and V is the one that is not. On the strength of the evidence: the proofs are elementary and every inequality is written out. Five numerical checks recompute quantities from the problem statement. Among them, the exact kernel at the origin is obtained by two independent routes that agree to 5×10^{-7}. The single audit round (Gemini, self-reported, with no code execution and no web access) returned ACCEPT with one NIT. It did not recompute the load-bearing divergence proofs, and two citations, Reed–Simon I Theorem VIII.31 and Simon 1982 Theorem B.7.1, remain unverified against the books.

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2. Setup

2.1 Conventions

Atomic units throughout: $\hbar = m_e = e^2/4\pi\varepsilon_0 = 1$. Lengths are in Bohr radii a_0; imaginary time and inverse temperature are in $\hbar/E_h$. $B_\varepsilon(\mathbf{x})$ is the open ball of radius ε centred at $\mathbf{x}$, and $B_\varepsilon = B_\varepsilon(\mathbf{0})$. $|A|$ is the Lebesgue measure of $A \subset \mathbb{R}^3$. "Kernel" means the integral kernel in the position representation.

2.2 Symbol table

Symbol	Meaning
$\mathbf{r} \in \mathbb{R}^3$, $r = \lvert\mathbf{r}\rvert$	position; distance from the nucleus
$K = -\frac{1}{2}\nabla^2$	kinetic energy operator on $L^2(\mathbb{R}^3)$
$V(\mathbf{r}) = -1/r$	attractive Coulomb potential
$H = K + V$	hydrogen Hamiltonian, infinitely heavy nucleus
$\beta > 0$	total imaginary time
$N \geq 1$	number of time slices (beads)
$\Delta\tau = \beta/N$	slice width
$\rho_0(\mathbf{r}, \mathbf{r}'; t)$	free kernel $(2\pi t)^{-3/2}\exp(-\lvert\mathbf{r}-\mathbf{r}'\rvert^2/2t)$, given in prompt 3a
$T = e^{-\Delta\tau K}e^{-\Delta\tau V}$	one-slice primitive operator
$w(\mathbf{r}) = e^{-\Delta\tau V(\mathbf{r})} = e^{\Delta\tau/r}$	potential weight of one bead
$\mathbf{r}_1, \ldots, \mathbf{r}_N$	bead positions, with $\mathbf{r}_{N+1} \equiv \mathbf{r}_1$
$Z_N(\beta)$, P_N	primitive trace defined by (A3), and its integrand
$Z_N^{(L)}$, $Z_N^{(L)}(\eta)$	(A3) with all beads in B_L; and additionally with $\lvert\mathbf{r}_j\rvert > \eta$
$\rho_N(\mathbf{r}, \mathbf{r}')$	primitive kernel $\langle\mathbf{r}\rvert T^N\lvert\mathbf{r}'\rangle$ written as a bead integral, (B10)
$I_a(\eta, \varepsilon)$	$4\pi\int_\eta^\varepsilon r^2 e^{a/r}\,dr$, with $a > 0$
$K_\tau(\mathbf{r}, \mathbf{r}')$	exact kernel $\langle\mathbf{r}\rvert e^{-\tau H}\lvert\mathbf{r}'\rangle$
$f_\delta(\mathbf{x})$	Kato-class quantity $\int_{B_\delta(\mathbf{x})} d^3y/(\lvert\mathbf{y}\rvert\lvert\mathbf{x}-\mathbf{y}\rvert)$
$\mu, \hbar, k, \Delta\beta, m$	the notes' mass, Planck constant, $k = e^2/4\pi\varepsilon_0$, slice width and number of slices

2.3 The problem restated

(a) Write $Z_N(\beta) = \mathrm{Tr}\,(e^{-\Delta\tau K}e^{-\Delta\tau V})^N$, with $\Delta\tau = \beta/N$, as an integral over N bead positions in $\mathbb{R}^3$, keeping every prefactor.

(b) Decide whether Z_N is finite for all $N \geq 1$ and $\Delta\tau > 0$, and prove the answer. If it is not, identify the configurations responsible and decide whether the kinetic factors can compensate.

(c) Decide whether $K_\tau(\mathbf{r}, \mathbf{r}')$ and $\mathrm{Tr}\,e^{-\beta H}$ are finite. State the theorem that justifies time slicing, and identify which of its hypotheses $-1/r$ violates.

The free kernel ρ_0 is given and trusted (prompt 3a). Everything in the notes is treated as a hypothesis, H1 to H6, and assessed in Part D.

2.4 Dictionary to the notes

The notes keep $\hbar$, μ and k. They write $\beta = \tau/\hbar$ (p. 4), $\Delta\beta = \beta/m$, and $\Delta\tau_{\text{notes}} = \hbar\Delta\beta$ (p. 7). The correspondence is:

- the prompt's $\Delta\tau$, an inverse energy, is the notes' $\Delta\beta$;
- the prompt's N is the notes' m;
- $\mu \to 1$, $\hbar \to 1$, $k \to 1$.

With $\hbar = 1$ the notes' $\Delta\tau$ and $\Delta\beta$ coincide numerically, as the notes say on p. 9. The notes' potential weight $w_{\text{pot}} = e^{\Delta\tau_{\text{notes}} k/(\hbar r)} = e^{\Delta\beta k/r}$ becomes $e^{\Delta\tau/r}$.

3. Derivation

Part A. The primitive trace as a bead integral

Write $Z_N = \operatorname{Tr} T^N$ with $T = e^{-\Delta\tau K} e^{-\Delta\tau V}$. Take the trace in the position representation and insert the resolution of the identity $1 = \int d^3r\, |\mathbf{r}\rangle\langle\mathbf{r}|$ between consecutive factors of T: [D]

$$Z_N = \int d^3r_1\, \langle \mathbf{r}_1 | T\, T \cdots T | \mathbf{r}_1 \rangle = \int d^3r_1 \cdots d^3r_N\, \langle \mathbf{r}_1|T|\mathbf{r}_N\rangle\langle \mathbf{r}_N|T|\mathbf{r}_{N-1}\rangle \cdots \langle \mathbf{r}_2|T|\mathbf{r}_1\rangle. \tag{A1}$$

With the label $\mathbf{r}_{N+1} \equiv \mathbf{r}_1$, the product is $\prod_{j=1}^{N} \langle \mathbf{r}_{j+1}|T|\mathbf{r}_j\rangle$.

Since V is a multiplication operator, $e^{-\Delta\tau V}|\mathbf{r}_j\rangle = e^{-\Delta\tau V(\mathbf{r}_j)}|\mathbf{r}_j\rangle$. Combining this with the given free kernel, [D]

$$\langle \mathbf{r}_{j+1}|e^{-\Delta\tau K}e^{-\Delta\tau V}|\mathbf{r}_j\rangle = e^{-\Delta\tau V(\mathbf{r}_j)}\rho_0(\mathbf{r}_{j+1},\mathbf{r}_j;\Delta\tau) = (2\pi\Delta\tau)^{-3/2} \exp\Big(-\frac{|\mathbf{r}_{j+1}-\mathbf{r}_j|^2}{2\Delta\tau}\Big)\, e^{\Delta\tau/r_j}. \tag{A2}$$

The potential is evaluated at the right-hand point of each link because V stands to the right of K in T.

Multiply the N factors (A2). The prefactors give $(2\pi\Delta\tau)^{-3N/2}$, and the product of exponentials is the exponential of the sum: [D]

$$Z_N(\beta) = (2\pi\Delta\tau)^{-3N/2} \int_{\mathbb{R}^{3N}} d^3r_1 \cdots d^3r_N\, \exp\Big\{-\sum_{j=1}^{N}\Big[\frac{|\mathbf{r}_{j+1}-\mathbf{r}_j|^2}{2\Delta\tau} - \frac{\Delta\tau}{r_j}\Big]\Big\}, \qquad \mathbf{r}_{N+1}\equiv\mathbf{r}_1, \quad \Delta\tau = \beta/N. \tag{A3}$$

Call the integrand P_N. The first term in the bracket is the discretised kinetic action, and the second is $\Delta\tau\, V(\mathbf{r}_j)$ with $V = -1/r$.

I take (A3) as the definition of Z_N. [A1]

- Why this is legitimate. P_N is a nonnegative measurable function on $\mathbb{R}^{3N}$. Its Lebesgue integral therefore exists in $[0, +\infty]$, and by Tonelli's theorem it equals every iterated integral, taken in any order. No exchange of limits is involved.
- Why a definition is needed. For this V, steps (A1) and (A2) are formal. Remark B.6 shows that T is unbounded and that T^N with $N \geq 2$ is defined only on the zero vector, so $\operatorname{Tr} T^N$ has no operator meaning here.
- What changes if it is wrong. If one insists on an operator trace, the answer to (b) becomes stronger: the object does not exist at all.
- For a bounded V, the same steps compose integral operators with nonnegative kernels and give the same formula.

In the notes' units (dictionary 2.4), the same formula reads [D]

$$Z_m = \Big(\frac{\mu}{2\pi\hbar^2\Delta\beta}\Big)^{3m/2} \int \prod_{j=1}^{m} d^3 r_j \, \exp\Big\{-\sum_{j=1}^{m}\Big[\frac{\mu|\mathbf{r}_{j+1}-\mathbf{r}_j|^2}{2\hbar^2\Delta\beta} + \Delta\beta\, V(\mathbf{r}_j)\Big]\Big\}, \qquad V = -k/r. \tag{A4}$$

Two properties of (A3) are used later.

- The bead labels are cyclic: P_N is unchanged when every $\mathbf{r}_j$ is replaced by $\mathbf{r}_{j+1}$ at once. So anything proved for bead 1 holds for every bead.
- Reversing the order of the factors in T (writing $e^{-\Delta\tau V}e^{-\Delta\tau K}$) moves the potential to the left-hand point of each link. After relabelling $j \to j+1$ the bead integral is the same, so (A3) does not depend on that choice.

Confidence for Part A: high. The only thing that would lower it is a different intended ordering or a symmetric splitting, and either leaves (A3) unchanged.

Part B. Z_N is infinite for every $N \geq 1$ and every $\Delta\tau > 0$

The claim is $Z_N(\beta) = +\infty$ for all $N \geq 1$ and $\Delta\tau > 0$. Two separate mechanisms produce it.

- Configurations with a bead near the origin. This one is specific to time slicing.
- The chain as a whole moving off to infinity. The exact trace shares this one.

Both are proved below, and then separated.

B.1 The one-bead radial integral

For $x \geq 0$, every term of $e^x = \sum_n x^n/n!$ is nonnegative. Keeping only the cubic term, [D]

$$e^x \geq \frac{x^3}{6} \qquad (x \geq 0). \tag{B1}$$

Take $a > 0$ and $0 < \eta < \varepsilon$. Apply (B1) with $x = a/r > 0$, multiply by r^2, and integrate: [D]

$$I_a(\eta,\varepsilon) = 4\pi \int_\eta^\varepsilon r^2 e^{a/r}\, dr \;\geq\; 4\pi \int_\eta^\varepsilon r^2\, \frac{a^3}{6r^3}\, dr = \frac{2\pi a^3}{3}\int_\eta^\varepsilon \frac{dr}{r} = \frac{2\pi a^3}{3}\ln\frac{\varepsilon}{\eta}. \tag{B2}$$

The right side tends to $+\infty$ as $\eta \to 0^+$, and $I_a(\eta,\varepsilon)$ increases as η decreases. By monotone convergence,

$$\int_{B_\varepsilon} e^{a/r}\, d^3 r = I_a(0,\varepsilon) = +\infty \qquad \text{for every } a > 0,\ \varepsilon > 0. \tag{B3}$$

The power 3 in (B1) is the smallest that works. The measure $r^2\,dr$ needs a factor r^{-3} to produce a logarithm, so any power $n \geq 3$ gives a divergence, while $n \leq 2$ would leave the bound inconclusive.

Now the growth law with an inner cutoff. Substitute $u = a/r$, so $r = a/u$, $dr = -a\,u^{-2}\,du$ [D] and $r^2\,dr = -a^3 u^{-4}\,du$. The limits $r = \eta$ and $r = \varepsilon$ become $U = a/\eta$ and $U_0 = a/\varepsilon$:

$$I_a(\eta,\varepsilon) = 4\pi a^3 \int_{U_0}^{U} u^{-4} e^u\, du. \tag{B4}$$

From $\frac{d}{du}(e^u u^{-n}) = e^u u^{-n} - n\,e^u u^{-n-1}$,

$$\int_{U_0}^{U} e^u u^{-n}\, du = \Big[e^u u^{-n}\Big]_{U_0}^{U} + n\int_{U_0}^{U} e^u u^{-n-1}\, du. \tag{B5}$$

Applying (B5) with $n = 4$, then $n = 5$, then $n = 6$:

$$\int_{U_0}^{U} e^u u^{-4}\,du = \Big[e^u\big(u^{-4} + 4u^{-5} + 20u^{-6}\big)\Big]_{U_0}^{U} + 120\int_{U_0}^{U} e^u u^{-7}\,du.$$

For $U > 2U_0$, split the last integral at $U/2$.

- On $[U_0, U/2]$, $e^u \le e^{U/2}$, so this piece is at most $e^{U/2}\int_{U_0}^{\infty} u^{-7}du = e^{U/2}U_0^{-6}/6$.
- On $[U/2, U]$, $u^{-7} \le (U/2)^{-7}$, so this piece is at most $2^7U^{-7}\int_{U/2}^{U} e^u\,du \le 2^7U^{-7}e^U$.

Both pieces are $O(e^U U^{-7})$ as $U \to \infty$; the first because $e^{U/2}$ grows more slowly than $e^U U^{-7}$. The boundary terms at U_0 do not depend on η. So, as $\eta \to 0$ with a and ε fixed,

$$I_a(\eta,\varepsilon) = 4\pi a^3 e^{a/\eta}\Big[\Big(\frac{\eta}{a}\Big)^4 + 4\Big(\frac{\eta}{a}\Big)^5 + 20\Big(\frac{\eta}{a}\Big)^6 + O\Big(\Big(\frac{\eta}{a}\Big)^7\Big)\Big], \qquad I_a(\eta,\varepsilon) \simeq \frac{4\pi\eta^4 e^{a/\eta}}{a}. \tag{B6}$$

One more application of (B5) gives 120 as the coefficient of $(\eta/a)^7$. The integrand $e^{a/r}$ drops by a factor e between $r = \eta$ and $r \approx \eta + \eta^2/a$, so the integral is dominated by a shell of thickness η^2/a at the cutoff. Check C1 tests (B2) and (B6) against direct quadrature.

B.2 Configurations with one bead near the origin

Fix $N \ge 1$, $\Delta\tau > 0$ and $0 < \varepsilon \le 1$, and define [D]

$$D_\varepsilon = \{\mathbf{r}_1 \in B_\varepsilon\} \times \{\mathbf{r}_k \in B_1,\ k = 2, \dots, N\} \subset \mathbb{R}^{3N}$$

(for $N = 1$, $D_\varepsilon = B_\varepsilon$). On D_ε every bead lies in B_1. By the triangle inequality, every link then satisfies $|\mathbf{r}_{k+1} - \mathbf{r}_k| \le |\mathbf{r}_{k+1}| + |\mathbf{r}_k| < 2$, including the closing link from $\mathbf{r}_N$ to $\mathbf{r}_{N+1} = \mathbf{r}_1$. Hence each kinetic factor obeys

$$\exp\Big(-\frac{|\mathbf{r}_{k+1} - \mathbf{r}_k|^2}{2\Delta\tau}\Big) \ \ge\ e^{-4/(2\Delta\tau)} = e^{-2/\Delta\tau} \qquad \text{on } D_\varepsilon. \tag{B7}$$

For $N = 1$ the only link has length $|\mathbf{r}_1 - \mathbf{r}_1| = 0$, so its factor equals 1, which also satisfies (B7). Since $V < 0$ everywhere, $w(\mathbf{r}_k) = e^{\Delta\tau/r_k} \ge 1$ for $k = 2, \dots, N$. Putting these together, on D_ε:

$$P_N \ \ge\ (2\pi\Delta\tau)^{-3N/2}\, e^{-2N/\Delta\tau}\, e^{\Delta\tau/r_1}. \tag{B8}$$

$P_N \ge 0$ everywhere, so the integral over $\mathbb{R}^{3N}$ is at least the integral over D_ε. Integrate the right side of (B8) over D_ε. The $N - 1$ beads confined to B_1 contribute $|B_1|^{N-1} = (4\pi/3)^{N-1}$, so

$$Z_N \ \ge\ (2\pi\Delta\tau)^{-3N/2}\, e^{-2N/\Delta\tau}\Big(\frac{4\pi}{3}\Big)^{N-1}\int_{B_\varepsilon} e^{\Delta\tau/r_1}\,d^3r_1 = +\infty \tag{B9}$$

by (B3) with $a = \Delta\tau$. This proves $Z_N = +\infty$ for every $N \ge 1$ and $\Delta\tau > 0$. By cyclic symmetry, any bead can play the role of bead 1.

The same argument works in localised forms. [D]

- Confining every bead to B_ℓ with $\ell = \min(L, 1)$ gives $Z_N^{(L)} = +\infty$ for every $L > 0$. What matters is only where the beads are relative to the origin. Any region of configuration space that contains a neighbourhood of the origin for one bead, with the other beads in a bounded set, carries infinite weight.
- For the primitive kernel with $N \ge 2$, define

$$\rho_N(\mathbf{r},\mathbf{r}') = \langle \mathbf{r}|T^N|\mathbf{r}'\rangle := (2\pi\Delta\tau)^{-3N/2}\int \prod_{j=2}^{N} d^3r_j \prod_{j=1}^{N} \exp\Big(-\frac{|\mathbf{r}_{j+1}-\mathbf{r}_j|^2}{2\Delta\tau}\Big) e^{\Delta\tau/r_j}, \qquad \mathbf{r}_1=\mathbf{r}',\ \mathbf{r}_{N+1}=\mathbf{r}. \tag{B10}$$

Restrict the intermediate bead $\mathbf{r}_2$ to B_ε and $\mathbf{r}_3,\dots,\mathbf{r}_N$ to B_1. Every link then has length at most $|\mathbf{r}|+|\mathbf{r}'|+2$, so every kinetic factor is bounded below by a positive constant that depends on $|\mathbf{r}|$ and $|\mathbf{r}'|$. Hence $\rho_N(\mathbf{r},\mathbf{r}') = +\infty$ for every pair of points when $N \geq 2$. - For $N = 1$, $\rho_1(\mathbf{r},\mathbf{r}') = \rho_0(\mathbf{r},\mathbf{r}';\Delta\tau)\, e^{\Delta\tau/r'}$ is finite for $\mathbf{r}' \neq \mathbf{0}$, but by (B3) it is not locally integrable in $\mathbf{r}'$ at $\mathbf{r}' = \mathbf{0}$.

B.3 Inside a bounded region, the origin is the only source

Take $0 < \delta < L$ and confine every bead to the shell $\delta < |\mathbf{r}_j| < L$. There each kinetic factor is at most 1 and each potential factor is at most $e^{\Delta\tau/\delta}$, so [D]

$$Z_N^{(L)}(\delta) \;\leq\; (2\pi\Delta\tau)^{-3N/2}\, e^{N\Delta\tau/\delta}\Big(\frac{4\pi L^3}{3}\Big)^N < \infty. \tag{B11}$$

So within any bounded region, the divergence comes entirely from configurations in which at least one bead approaches the origin, and B.2 shows that one such bead is enough.

B.4 The chain moving off to infinity

First, how free kernels compose. For one Cartesian component, with $s,t>0$, complete the square in y: [D]

$$-\frac{(x-y)^2}{2s} - \frac{(y-z)^2}{2t} = -\frac{s+t}{2st}\Big(y - \frac{tx+sz}{s+t}\Big)^2 - \frac{(x-z)^2}{2(s+t)}.$$

Integrate over $y \in \mathbb{R}$, using $\int \exp(-(s+t)(y-c)^2/2st)\, dy = \big(2\pi st/(s+t)\big)^{1/2}$:

$$\int_{\mathbb{R}} \frac{e^{-(x-y)^2/2s}}{\sqrt{2\pi s}}\,\frac{e^{-(y-z)^2/2t}}{\sqrt{2\pi t}}\,dy = \frac{1}{\sqrt{2\pi s}\sqrt{2\pi t}}\sqrt{\frac{2\pi st}{s+t}}\, e^{-(x-z)^2/2(s+t)} = \frac{e^{-(x-z)^2/2(s+t)}}{\sqrt{2\pi(s+t)}}.$$

The three-dimensional kernel is a product of three such factors, so

$$\int_{\mathbb{R}^3} \rho_0(\mathbf{x},\mathbf{y};s)\,\rho_0(\mathbf{y},\mathbf{z};t)\,d^3y = \rho_0(\mathbf{x},\mathbf{z};s+t). \tag{B12}$$

Check C3 tests (B12).

Next, a lower bound by the free chain. Since $e^{\Delta\tau/r_j} \geq 1$, [D]

$$P_N \geq (2\pi\Delta\tau)^{-3N/2}\exp\Big(-\sum_j \frac{|\mathbf{r}_{j+1}-\mathbf{r}_j|^2}{2\Delta\tau}\Big) = \prod_j \rho_0(\mathbf{r}_{j+1},\mathbf{r}_j;\Delta\tau).$$

Integrate out the beads one at a time with (B12), using the symmetry $\rho_0(\mathbf{x},\mathbf{y}) = \rho_0(\mathbf{y},\mathbf{x})$:

- over $\mathbf{r}_2$: $\int \rho_0(\mathbf{r}_3,\mathbf{r}_2;\Delta\tau)\rho_0(\mathbf{r}_2,\mathbf{r}_1;\Delta\tau)\,d^3r_2 = \rho_0(\mathbf{r}_3,\mathbf{r}_1;2\Delta\tau)$;
- over $\mathbf{r}_3$: $\int \rho_0(\mathbf{r}_4,\mathbf{r}_3;\Delta\tau)\rho_0(\mathbf{r}_3,\mathbf{r}_1;2\Delta\tau)\,d^3r_3 = \rho_0(\mathbf{r}_4,\mathbf{r}_1;3\Delta\tau)$;
- and so on. After $N-1$ steps what remains is $\rho_0(\mathbf{r}_{N+1},\mathbf{r}_1;N\Delta\tau) = \rho_0(\mathbf{r}_1,\mathbf{r}_1;\beta) = (2\pi\beta)^{-3/2}$. For $N=1$ no integration is needed.

Hence

$$Z_N \;\geq\; \int_{\mathbb{R}^3} (2\pi\beta)^{-3/2}\, d^3 r_1 = +\infty. \tag{B13}$$

This divergence has nothing to do with $r = 0$. It is present for $V \equiv 0$, where $Z_N = \operatorname{Tr} e^{-\beta K} = \infty$, and it comes from the region where all beads are far from the origin: translating a compact chain to infinity costs nothing. Part C shows the exact trace diverges for the same reason.

The problem asks for "the region responsible". There are two.

- The neighbourhoods $\{\mathbf{r}_j \in B_\varepsilon\}$ of the origin cause the failure specific to time slicing (B.2), and they are the only source inside bounded regions (B.3).
- The region at infinity produces a second divergence, which the continuum shares (B.4 and C.4).

B.5 The kinetic factors cannot compensate

Three statements, from weakest to strongest.

First, the kinetic weight satisfies $\prod_j \rho_0(\mathbf{r}_{j+1}, \mathbf{r}_j; \Delta\tau) \leq (2\pi\Delta\tau)^{-3N/2}$, with equality exactly [D] when all beads coincide. In particular it is maximal on the fully collapsed configuration $\mathbf{r}_1 = \cdots = \mathbf{r}_N = \mathbf{0}$, which is where the potential weight is most singular. The kinetic factor penalises the separation between neighbouring beads. It does not penalise a bead's distance from the origin.

Second, for one bead's integral to converge at the origin, the kinetic factor would have to [D] vanish as $\mathbf{r}_j \to \mathbf{0}$ fast enough to make $r_j^2 e^{\Delta\tau/r_j} \times$ (kinetic factor) integrable. But the kinetic factor depends on $\mathbf{r}_j$ only through $\mathbf{r}_j - \mathbf{r}_{j\pm1}$, is continuous, and is strictly positive at $\mathbf{r}_j = \mathbf{0}$ for fixed neighbours. That is what (B7) to (B9) express.

Third, the kinetic factors do not even change the leading growth. Impose the inner cutoff η [D] on every bead and confine all beads to B_L, with $0 < \eta < L$. Every kinetic factor is at most 1, so

$$Z_N^{(L)}(\eta) \;\leq\; (2\pi\Delta\tau)^{-3N/2}\, I_{\Delta\tau}(\eta, L)^N. \tag{B14}$$

For a lower bound, fix $\theta > 0$ with $\eta(1+\theta) < L$, and keep only configurations with every bead in the thin shell $\eta < |\mathbf{r}_j| < \eta(1+\theta)$. There each link has length at most $2\eta(1+\theta)$, so the kinetic exponent is at least $-N\big(2\eta(1+\theta)\big)^2/(2\Delta\tau) = -2N\eta^2(1+\theta)^2/\Delta\tau$. The potential factors then integrate bead by bead:

$$Z_N^{(L)}(\eta) \;\geq\; (2\pi\Delta\tau)^{-3N/2}\, e^{-2N\eta^2(1+\theta)^2/\Delta\tau}\, I_{\Delta\tau}\big(\eta, \eta(1+\theta)\big)^N. \tag{B15}$$

To compare the two bounds, write $I(\eta, \eta(1+\theta)) = I(\eta, L) - I(\eta(1+\theta), L)$ and bound each piece.

- $I(\eta(1+\theta), L) \leq (4\pi/3) L^3 e^{\Delta\tau/(\eta(1+\theta))}$: the integrand is at most its value at the inner end, and the volume is at most $|B_L|$.
- $I(\eta, L) \geq 4\pi \int_\eta^{\eta(1+\theta/2)} r^2 e^{\Delta\tau/r}\, dr \geq 4\pi\eta^2(\eta\theta/2)\, e^{\Delta\tau/(\eta(1+\theta/2))}$.

So

$$\frac{I(\eta(1+\theta), L)}{I(\eta, L)} \;\leq\; \frac{2L^3}{3\theta\eta^3} \exp\Big[-\frac{\Delta\tau}{\eta}\Big(\frac{1}{1+\theta/2} - \frac{1}{1+\theta}\Big)\Big] \;\longrightarrow\; 0 \quad (\eta \to 0),$$

because the bracket is positive and the exponential beats any power of η. Combining (B14), (B15) and $e^{-2N\eta^2(1+\theta)^2/\Delta\tau} \to 1$,

$$\lim_{\eta\to 0}\frac{Z_N^{(L)}(\eta)}{(2\pi\Delta\tau)^{-3N/2}I_{\Delta\tau}(\eta,L)^N}=1,\qquad Z_N^{(L)}(\eta)\simeq(2\pi\Delta\tau)^{-3N/2}\Big(\frac{4\pi\eta^4e^{\Delta\tau/\eta}}{\Delta\tau}\Big)^N. \tag{B16}$$

The divergence is the N-th power of the one-bead divergence, exactly as if every kinetic factor were replaced by its maximum.

For $N=2$ there is a sharper prediction, which check C2 tests. [D]

- The geometry. At $N=2$ the two links put $|\mathbf{r}_2-\mathbf{r}_1|^2/\Delta\tau$ in the exponent. With radii r_1,r_2 and relative angle θ_{12}, $|\mathbf{r}_1-\mathbf{r}_2|^2=r_1^2+r_2^2-2r_1r_2\cos\theta_{12}$. Integrating $\hat{\mathbf{r}}_2$ over the sphere with polar axis $\hat{\mathbf{r}}_1$ gives $2\pi\int_{-1}^{1}dc$; integrating $\hat{\mathbf{r}}_1$ then gives 4π; and $\int_{-1}^{1}e^{\kappa c}dc=2\sinh\kappa/\kappa$ with $\kappa=2r_1r_2/\Delta\tau$. So

$$\int d\Omega_1\,d\Omega_2\,e^{-|\mathbf{r}_1-\mathbf{r}_2|^2/\Delta\tau}=16\pi^2\,e^{-(r_1^2+r_2^2)/\Delta\tau}\,\frac{\sinh\kappa}{\kappa}. \tag{B17}$$

- The ratio. Let $g(r_1,r_2)$ be (B17) divided by $16\pi^2$, the angular average of the kinetic factor, and let $R(\eta)=Z_2^{(L)}(\eta)/\big[(2\pi\Delta\tau)^{-3}I(\eta,L)^2\big]$. Then $R=\iint\tilde w(r_1)\tilde w(r_2)\,g\,/\big(\int\tilde w\big)^2$ with $\tilde w(r)=r^2e^{\Delta\tau/r}$ on $[\eta,L]$, and since $g\le 1$, $R\le 1$.
- The deviation. With both radii in the shell, $e^{-s}=1-s+O(s^2)$ and $\sinh\kappa/\kappa=1+\kappa^2/6+O(\kappa^4)$ give $1-g=(r_1^2+r_2^2)/\Delta\tau+O(\eta^4/\Delta\tau^2)$. In the shell the radial weight is approximately $e^{\Delta\tau/\eta}e^{-\Delta\tau(r-\eta)/\eta^2}$, an exponential with mean width $\eta^2/\Delta\tau$, so $\langle r^2\rangle\approx\eta^2+2\eta\cdot\eta^2/\Delta\tau=\eta^2(1+2\eta/\Delta\tau)$. Pairs with a bead outside the shell are suppressed by the vanishing ratio shown above.

Hence

$$1-R(\eta)\;\simeq\;\frac{2\eta^2}{\Delta\tau}\Big(1+\frac{2\eta}{\Delta\tau}+\dots\Big),\qquad R(\eta)\to 1^-. \tag{B18}$$

If the kinetic factor compensated the divergence, R would go to 0 instead.

B.6 Remark: the divergence is not an artifact of the bead representation

Let M be multiplication by $w(r)=e^{\Delta\tau/r}$ on its maximal domain $D(M)=\{\psi\in L^2:\ w\psi\in L^2\}$, and let $T=e^{-\Delta\tau K}M$ with $D(T)=D(M)$. The claim is that $D(T^N)=\{0\}$ for $N\ge 2$, and that T is unbounded. [D]

Proof that $D(T^2)=\{0\}$.

1. Let $\psi\in D(T^2)$. Then $g:=w\psi\in L^2$, and $\varphi:=T\psi=e^{-\Delta\tau K}g$ must lie in $D(M)$.
2. φ is real-analytic. It is the convolution of g with the Gaussian $\rho_0(\cdot;\Delta\tau)$, and it extends to an entire function on $\mathbb{C}^3$. For $\mathbf{z}=\mathbf{x}+i\boldsymbol{\xi}$,

$$\Big|\exp\Big(-\sum_i(z_i-y_i)^2/2\Delta\tau\Big)\Big|=\exp\Big(-(|\mathbf{x}-\mathbf{y}|^2-|\boldsymbol{\xi}|^2)/2\Delta\tau\Big).$$

   Multiplied by $|g(\mathbf{y})|$, this is integrable in $\mathbf{y}$ (by Cauchy–Schwarz, the Gaussian being in L^2), locally uniformly in $\mathbf{z}$. Differentiating under the integral sign then shows holomorphy in each variable.
3. Suppose $\varphi\not\equiv 0$. A real-analytic function on a connected open set that vanishes to infinite order at one point vanishes identically [R]. So the Taylor series of φ at $\mathbf{0}$ has a lowest nonzero homogeneous part P_k, of some degree $k\ge 0$.
4. P_k cannot vanish on the whole unit sphere, since by homogeneity it would then vanish everywhere. So there is a spherical cap S_0 of positive area σ_0 on which $|P_k|\ge c>0$.

5. Taylor's theorem gives $|\varphi(\mathbf{x}) - P_k(\mathbf{x})| \le C|\mathbf{x}|^{k+1}$ for $|\mathbf{x}| \le r_0$. For $\mathbf{x} = s\boldsymbol{\omega}$ with $\boldsymbol{\omega} \in S_0$ and $s \le s_1 := \min(r_0, c/2C)$, this gives $|\varphi(\mathbf{x})| \ge cs^k - Cs^{k+1} \ge cs^k/2$.
6. Using $e^x \ge x^{2k+3}/(2k+3)!$ with $x = 2\Delta\tau/s$,

$$\int_{B_1} w^2|\varphi|^2\, d^3x \;\ge\; \sigma_0 \int_0^{s_1} s^2 e^{2\Delta\tau/s}\, \frac{c^2 s^{2k}}{4}\, ds \;\ge\; \frac{\sigma_0 c^2}{4}\, \frac{(2\Delta\tau)^{2k+3}}{(2k+3)!} \int_0^{s_1} \frac{ds}{s} = +\infty.$$

   This contradicts $\varphi \in D(M)$, so $\varphi \equiv 0$.
7. In Fourier space $e^{-\Delta\tau K}$ is multiplication by $e^{-\Delta\tau|\mathbf{k}|^2/2} > 0$, so it is injective, and $g = 0$. Since $w > 0$, $\psi = 0$ almost everywhere.
8. Finally, $D(T^N) \subset D(T^2)$ for every $N \ge 2$.

Proof that T is unbounded. Take $\psi \ge 0$ supported in B_1. Since $|\mathbf{x} - \mathbf{y}| \le |\mathbf{x}| + 1$ on the support,

$$(T\psi)(\mathbf{x}) \ge (2\pi\Delta\tau)^{-3/2} e^{-(|\mathbf{x}|+1)^2/2\Delta\tau} \int w\psi\, d^3y,$$

so $\|T\psi\|_2 \ge c_{\Delta\tau} \int w\psi$ with $c_{\Delta\tau} > 0$. Now let ψ_n be the normalised indicator of the shell $A_n = \{1/(2n) < |\mathbf{y}| < 1/n\}$. It lies in $D(T)$, and $\int w\psi_n \ge e^{\Delta\tau n}|A_n|^{1/2}$ with $|A_n| = \frac{7}{8} \cdot \frac{4\pi}{3} n^{-3}$, which tends to ∞. So T is unbounded, and in particular not trace class.

Confidence for Part B:

- high for B.1 to B.5, which are elementary, with every inequality written out, and agree with checks C1 to C3;
- medium-high for B.6, which uses the identity theorem for real-analytic functions [R] and is not checked numerically.

Part C. The continuum

C.1 H is self-adjoint

Split $V = V_1 + V_2$ with $V_1 = -\chi_{r<1}/r$ and $V_2 = -\chi_{r\ge1}/r$. [V]

- $\|V_1\|_2^2 = \int_{r<1} r^{-2}\, d^3r = 4\pi \int_0^1 dr = 4\pi < \infty$.
- $|V_2| \le 1$.

So $V \in L^2(\mathbb{R}^3) + L^\infty(\mathbb{R}^3)$. Such a V is $-\Delta$-bounded with relative bound zero (Simon 2017, §7, the discussion after eq. (7.15)). The Kato–Rellich theorem (same source, Theorem 7.3) then makes $H = K + V$ self-adjoint on $D(K) = H^2(\mathbb{R}^3)$ and bounded below. In particular $D(K) \subset D(V)$.

C.2 $-1/r$ is in the Kato class

In three dimensions the Kato class K_3 consists of the V with $\lim_{\delta\to0} \sup_{\mathbf{x}} \int_{|\mathbf{x}-\mathbf{y}|\le\delta} |V(\mathbf{y})|\, |\mathbf{x} - \mathbf{y}|^{-1}\, d^3y = 0$ (Shen 2001, eq. (1.1)). [V]

For $|V(\mathbf{y})| = 1/|\mathbf{y}|$ the quantity to control is $f_\delta(\mathbf{x}) = \int_{B_\delta(\mathbf{x})} d^3y/(|\mathbf{y}||\mathbf{x} - \mathbf{y}|)$. For $a, b > 0$, $(1/a - 1/b)^2 \ge 0$ gives $1/(ab) \le (1/a^2 + 1/b^2)/2$, so [D]

$$f_\delta(\mathbf{x}) \le \frac{1}{2} \int_{B_\delta(\mathbf{x})} \frac{d^3y}{|\mathbf{y}|^2} + \frac{1}{2} \int_{B_\delta(\mathbf{x})} \frac{d^3y}{|\mathbf{x} - \mathbf{y}|^2}. \tag{C1}$$

The second integral equals $\int_{B_\delta} |\mathbf{z}|^{-2} d^3z = 4\pi \int_0^\delta dr = 4\pi\delta$. For the first, let $A = B_\delta(\mathbf{x})$, $B = B_\delta(\mathbf{0})$ and $\phi(\mathbf{y}) = |\mathbf{y}|^{-2}$.

- Since $|A| = |B|$, also $|A \setminus B| = |B \setminus A|$.

- On $A \setminus B$, $|\mathbf{y}| \geq \delta$, so $\phi \leq \delta^{-2}$.
- On $B \setminus A$, $|\mathbf{y}| < \delta$, so $\phi > \delta^{-2}$.

Hence

$$\int_A \phi - \int_B \phi = \int_{A\setminus B} \phi - \int_{B\setminus A} \phi \;\leq\; \delta^{-2}|A \setminus B| - \delta^{-2}|B \setminus A| = 0, \tag{C2}$$

so the first integral is at most $\int_B \phi = 4\pi\delta$. Together,

$$\sup_{\mathbf{x}} f_\delta(\mathbf{x}) \;\leq\; 4\pi\delta \;\longrightarrow\; 0 \quad (\delta \to 0), \tag{C3}$$

so $-1/r \in K_3$, and its positive part is zero. Check C5 computes f_δ by quadrature; the bound is attained at $\mathbf{x} = \mathbf{0}$, where $f_\delta(\mathbf{0}) = 4\pi\delta$.

C.3 The exact kernel is finite, including at the origin

If $V_- \in K_\nu$ and $V_+ \in K_\nu^{\mathrm{loc}}$, then e^{-tH} has a jointly continuous integral kernel for every $t > 0$. [S]
This is Simon 1982, Theorem B.7.1, {ver: no}. Broderix, Hundertmark and Leschke (2000) prove the same for Kato-class scalar potentials with magnetic fields; vector potential $A = 0$ on $\Omega = \mathbb{R}^3$ is the case Simon treated (abstract and §6, {ver: tool}). By C.2 the theorem applies, so $K_\tau(\mathbf{r}, \mathbf{r}')$ is finite and continuous on $\mathbb{R}^3 \times \mathbb{R}^3$ for every $\tau > 0$, including at $\mathbf{r} = \mathbf{r}' = \mathbf{0}$.

I also evaluate the kernel explicitly at the origin, which is where the primitive kernel fails. [D, from recall inputs]
Only $l = 0$ states are nonzero at $r = 0$.

- Bound states. The $l = 0$ bound states have $E_n = -1/(2n^2)$ [R] and $|\psi_{n00}(0)|^2 = 1/(\pi n^3)$. [V] Fäldt and Wilkin give $[\psi_n(0)]^2 = 4\alpha_n^3$ in the radial normalisation $\psi = u/r$, $\int u^2 dr = 1$ (their eq. (21)). In atomic units $\alpha_n = 1/n$, so $u_n'(0)^2 = 4/n^3$, and the full wavefunction $\psi_{n00} = u_n/(\sqrt{4\pi}\, r)$ gives $4/(4\pi n^3) = 1/(\pi n^3)$.
- Continuum. Normalise the scattering states to $(2\pi)^{-3/2} e^{i\mathbf{k}\cdot\mathbf{r}}$ at infinity. Their value at the origin is the free value $(2\pi)^{-3/2}$ times the Sommerfeld factor. [V] Fäldt and Wilkin give $[\psi(k,0)]^2 = 2\pi\eta/(e^{2\pi\eta} - 1)$, with $\eta = -me^2/(\hbar^2 k)$ in the attractive case, normalised so that the free value is 1 (their eq. (19)). In atomic units $\eta = -1/k$, so

$$C^2(k) = \frac{2\pi(-1/k)}{e^{-2\pi/k} - 1} = \frac{2\pi/k}{1 - e^{-2\pi/k}}. \tag{C4}$$

Assuming completeness of the bound states together with the Coulomb waves [R],

$$K_\tau(\mathbf{0}, \mathbf{0}) = \sum_{n\geq 1} \frac{e^{\tau/2n^2}}{\pi n^3} + \int \frac{d^3k}{(2\pi)^3}\, C^2(k)\, e^{-\tau k^2/2} = \sum_{n\geq 1} \frac{e^{\tau/2n^2}}{\pi n^3} + \frac{1}{\pi}\int_0^\infty \frac{k\, e^{-\tau k^2/2}}{1 - e^{-2\pi/k}}\, dk. \tag{C5}$$

The second form uses $d^3k = 4\pi k^2 dk$ and $k^2 C^2/(2\pi^2) = k/\big(\pi(1 - e^{-2\pi/k})\big)$.
Both terms are finite for every $\tau > 0$.

- The sum is bounded term by term (step added in round 2 for finding F1.1). For $n \geq 1$, [D]
$1/n^2 \leq 1$; since $\tau > 0$, this gives $\tau/(2n^2) \leq \tau/2$, and because the exponential is increasing, $e^{\tau/2n^2} \leq e^{\tau/2}$. Hence

$$\sum_{n\geq 1} \frac{e^{\tau/2n^2}}{\pi n^3} \;\leq\; \frac{e^{\tau/2}}{\pi} \sum_{n\geq 1} \frac{1}{n^3} = \frac{e^{\tau/2}\, \zeta(3)}{\pi}.$$

Here $\zeta(3) = \sum_{n\geq 1} n^{-3} \approx 1.2021$ is finite, being a p-series with $p = 3 > 1$. The bound is sharp as $\tau \to 0$, where both sides tend to $\zeta(3)/\pi$. Check C6 in §10 tests it.

- The integrand behaves like k as $k \to 0$, and like $k^2 e^{-\tau k^2/2}/(2\pi)$ as $k \to \infty$.
- Consistency: replacing C^2 by 1 and dropping the bound states gives $(1/2\pi^2)\int_0^\infty k^2 e^{-\tau k^2/2} dk = (2\pi\tau)^{-3/2}$, the free value.

Check C4 evaluates (C5) and compares it with an independent finite-difference diagonalisation of the $l = 0$ radial Hamiltonian, which uses none of the recalled data. After extrapolation in the grid spacing the two agree to better than 5×10^{-7} relative, for $\tau = 0.5, 1, 2, 4$. At $\tau = 1$, $K_1(\mathbf{0}, \mathbf{0}) = 0.9195694112$, which is 14.48 times the free value 0.0634936359. [N]

C.4 The exact trace is infinite

Step 1, a Jensen inequality from the spectral theorem. For a unit vector $\varphi \in D(H)$, the spectral measure μ_φ of H is a probability measure with $\langle \varphi, f(H)\varphi \rangle = \int f \, d\mu_\varphi$ and $m := \langle \varphi, H\varphi \rangle = \int \lambda \, d\mu_\varphi$ [R, functional calculus]. By convexity of $\lambda \mapsto e^{-\beta\lambda}$, $e^{-\beta\lambda} \geq e^{-\beta m}(1 - \beta(\lambda - m))$ for every real λ. Integrating against μ_φ: [D]

$$\langle \varphi, e^{-\beta H}\varphi \rangle \;\geq\; e^{-\beta m}\Big(1 - \beta\Big(\int \lambda \, d\mu_\varphi - m\Big)\Big) = e^{-\beta\langle \varphi, H\varphi \rangle}. \tag{C6}$$

Both integrals exist: $e^{-\beta\lambda}$ is bounded on the spectrum because H is bounded below, and λ is μ_φ-integrable because $\varphi \in D(H)$.

Step 2, infinitely many orthogonal test states. Take $\varphi \in C_c^\infty(\mathbb{R}^3)$ with $\|\varphi\| = 1$ and support in B_1, and set $\varphi_k(\mathbf{r}) = \varphi(\mathbf{r} - 3k\hat{\mathbf{e}}_z)$ for $k = 0, 1, 2, \ldots$. [D]

- The supports are disjoint, so the φ_k are orthonormal.
- K commutes with translations and $V \leq 0$, so $\langle \varphi_k, H\varphi_k \rangle \leq \langle \varphi, K\varphi \rangle =: c < \infty$.
- For a positive operator, the trace is at least the sum of the diagonal elements over any orthonormal set: extend the set to a basis, and every added term is nonnegative.

So, by (C6),

$$\operatorname{Tr} e^{-\beta H} \;\geq\; \sum_{k \geq 0} \langle \varphi_k, e^{-\beta H}\varphi_k \rangle \;\geq\; \sum_{k \geq 0} e^{-\beta c} = +\infty \qquad \text{for every } \beta > 0. \tag{C7}$$

The bound states alone give the same conclusion [R]: they contribute $\sum_n n^2 e^{\beta/2n^2} \geq \sum_n n^2 = \infty$. The argument through (C7) shows that the divergence is a large-volume effect: infinitely many disjoint regions of space each contribute at least $e^{-\beta c}$. Over a bounded region the exact diagonal kernel integrates to a finite number, $\int_{B_L} K_\beta(\mathbf{r}, \mathbf{r})\, d^3r < \infty$, because K_β is continuous (C.3).

C.5 Comparison

Quantity	Primitive, any $N \geq 1$, $\Delta\tau > 0$	Exact
Full trace over $\mathbb{R}^3$	$+\infty$, from two sources: a bead near the origin (B9) and the chain at infinity (B13)	$+\infty$, from the large-volume source only (C7)
Diagonal kernel integrated over B_L: $\int_{B_L} \rho_N(\mathbf{r}, \mathbf{r}) d^3r$ vs $\int_{B_L} K_\beta(\mathbf{r}, \mathbf{r}) d^3r$	$+\infty$ for every $L > 0$ (B.2)	finite (C.3)

Quantity	Primitive, any $N \geq 1$, $\Delta\tau > 0$	Exact
Kernel at fixed points $(\mathbf{r}, \mathbf{r}')$	$+\infty$ at every pair when $N \geq 2$ (B10); for $N = 1$, finite if $\mathbf{r}' \neq \mathbf{0}$ but not locally integrable at $\mathbf{r}' = \mathbf{0}$	finite and continuous everywhere, e.g. $K_1(\mathbf{0}, \mathbf{0}) = 0.91957$ (check C4)

The large-volume divergence is common to both sides and has nothing to do with $r = 0$. The divergence at the origin occurs in every primitive approximant and has no counterpart in the continuum. Since $Z_N^{(L)} = \infty$ for every N, the localised primitive trace cannot converge to the finite localised exact quantity as $N \to \infty$. The failure happens at every finite N, not only in the limit.

C.6 The theorem that justifies time slicing, and the hypothesis $-1/r$ violates

The Trotter product formula, in the form given by Reed and Simon. Let A and B be self-adjoint [v]
on a Hilbert space, with $A + B$ essentially self-adjoint on $D(A) \cap D(B)$. Then for real t,

$$e^{-it\overline{(A+B)}} = \operatorname*{s-lim}_{n\to\infty} \left(e^{-itA/n} e^{-itB/n}\right)^n. \tag{C8}$$

If, in addition, A and B are bounded below, then for $t \geq 0$,

$$e^{-t\overline{(A+B)}} = \operatorname*{s-lim}_{n\to\infty} \left(e^{-tA/n} e^{-tB/n}\right)^n. \tag{C9}$$

What was retrieved:

- the unitary form, with the essential-self-adjointness hypothesis, as Reed–Simon I, Theorem VIII.31 (quoted in arXiv:2501.04564, Theorem B.3.7);
- the semigroup form, with "A and B bounded from below" (Encyclopedia of Mathematics, "Trotter product formula"; arXiv:2308.00290, Theorem A1.17, which attributes it to Teschl).

I have not read the semigroup clause in Reed–Simon itself: {ver: no}.
The hypotheses for $A = K$ and $B = V$, applied to Z_N with $t = \beta$ and $n = N$:

Hypothesis	Status for K, V	Why
(i) A and B self-adjoint	holds	K on $H^2(\mathbb{R}^3)$; V as a real multiplication operator on its maximal domain
(ii) $A + B$ essentially self-adjoint on $D(A) \cap D(B)$	holds	by C.1, $D(K) \subset D(V)$, so $D(K) \cap D(V) = D(K)$, and H is self-adjoint there
(iii) A and B bounded below	fails for V	$K \geq 0$, but $\inf_r(-1/r) = -\infty$

So the real-time formula (C8) holds for hydrogen, $e^{-itH} = \text{s-lim}\left(e^{-itK/N} e^{-itV/N}\right)^N$, because $e^{-itV/N}$ is unitary whatever the sign of V. The imaginary-time formula (C9), which is what (A3) relies on, is not available.

This is more than a missing step in a proof. With V unbounded below:

- $e^{-\Delta\tau V}$ is an unbounded operator (its norm would be $\sup_r e^{\Delta\tau/r} = \infty$);
- the approximants $\left(e^{-\Delta\tau K} e^{-\Delta\tau V}\right)^N$ are not bounded operators, and for $N \geq 2$ they are defined only on the zero vector (Remark B.6);

- so the strong limit in (C9) cannot even be formulated.

The hypothesis is about the summand V, not the sum: H itself is bounded below. Kato's 1978 extension of the product formula removes the domain condition (ii) by working with form sums. It still requires A and B to be positive self-adjoint operators, that is, generators of self-adjoint contraction semigroups (Simon 2017, §18, Theorem 18.2). $-1/r$ violates that requirement too, and no other.

Why the continuum survives when slicing fails, qualitatively. In the Feynman–Kac representation, the potential enters through the time integral $\int_0^\tau V(\mathbf{b}(s))\,ds$ along Brownian paths. For $V \in K_3$ this integral has finite exponential moments (Simon 1982, Khas'minskii's lemma, {ver: no}). The primitive weight replaces it by the Riemann sum $\Delta\tau \sum_j V(\mathbf{r}_j)$, which samples V at isolated instants, and the weight $e^{\Delta\tau/r_j}$ of a single sample is not locally integrable. This is offered as diagnosis only and is not used anywhere above. [R]

Confidence for Part C:

- high for C.1, C.2 and C.4, and for the kernel at the origin, where two independent routes agree;
- medium-high for the kernel at general points, which rests on a theorem whose locator I could not open (Simon 1982, B.7.1), supported by the retrieved abstract of Broderix, Hundertmark and Leschke;
- medium-high for the exact wording of the hypotheses in C.6, which I verified only through secondary sources. The identification of the violated hypothesis does not depend on whether the clause reads "bounded below" or "positive", because V is neither.

Part D. Assessment of the notes' hypotheses H1 to H6

H1 (p. 4) has two parts.

- Correct: the trace closes the paths, $x(\beta) = x(0)$. This is the $\mathbf{r}_{N+1} = \mathbf{r}_1$ of (A3).
- Wrong for this Hamiltonian: the ground-state dominance $Z(\beta) = \sum_n e^{-\beta E_n} \simeq e^{-\beta E_0}$ and the formula $E_0 \approx -\ln Z(\beta)/\beta$. Both presuppose that $e^{-\beta H}$ is trace class with a discrete spectrum. For hydrogen in $\mathbb{R}^3$, $\operatorname{Tr} e^{-\beta H} = +\infty$ for every β (C7). The bound-state sum alone diverges, and the continuum $[0, \infty)$ adds an infinite contribution. Read literally, the formula gives $E_0 \approx -\infty$.

Ground-state dominance does hold for the diagonal kernel $\langle \mathbf{r}|e^{-\beta H}|\mathbf{r}\rangle$ at fixed $\mathbf{r}$, which is finite (C.3) [R, not proved here]. That is a statement about a kernel, not about Z.

H2 (p. 5) is wrong for this V. As a justification of time slicing it would be incomplete even for matrices.

- For bounded operators, the per-slice agreement $e^{\Delta A}e^{\Delta B} = e^{\Delta(A+B)} + O(\Delta^2)$ is true. But the global statement needs the telescoping bound $\|X^N - Y^N\| \le N \max(\|X\|, \|Y\|)^{N-1}\|X - Y\|$, which turns N errors of order Δ^2 into $O(1/N)$ and requires uniform bounds on $\|e^{\Delta A}\|$ and $\|e^{\Delta B}\|$. The notes stop at the per-slice expansion.
- For $V = -1/r$, the expansion $e^{-\Delta\beta V} \approx 1 - \Delta\beta V$ fails at every $r \lesssim \Delta\beta$ (in atomic units), however small $\Delta\beta$ is, because $\Delta\beta V$ is not uniformly small in r. The left side $e^{-\Delta\beta(K+V)}$ is a bounded operator (H is bounded below). The right side $e^{-\Delta\beta K}e^{-\Delta\beta V}$ is unbounded (Remark B.6). So the two are not close in any operator sense, for any $\Delta\beta > 0$.

The notes' phrase "valid only for very small intervals" is exactly what fails: near the nucleus, no interval is small enough. The justification that does exist is the Trotter theorem, and its semiboundedness hypothesis is the one V violates (C.6).

H3 (p. 6) is wrong as written, in two substantive places and two minor ones.

- Wrong potential weight. The potential term reads $\beta V(x_j)$ where it should read $\Delta\beta V(x_j)$. The first line of p. 6 has $e^{-\Delta\beta V(x_j)}$, which is correct; the Δ is lost in the second line and stays lost through the final formula. Page 7 then writes S_m with $\Delta\tau\, V(x_j)$ and $\Delta\tau = \hbar\Delta\beta$, which is the correct per-slice weight, so the error is confined to p. 6. With βV the potential is overweighted by a factor m. The qualitative answer to (b) survives, since $e^{\beta/r}$ is no more integrable than $e^{\Delta\beta/r}$, but the growth law (B6) would have $a = \beta$ instead of $\Delta\beta$.
- One-dimensional formula. The prefactor $(\mu/2\pi\hbar^2\Delta\beta)^{m/2}$ and the scalar x_j describe a particle on a line. For hydrogen in $\mathbb{R}^3$ the prefactor is $(\mu/2\pi\hbar^2\Delta\beta)^{3m/2}$, the measures are d^3r_j, and the kinetic term is $|\mathbf{r}_{j+1} - \mathbf{r}_j|^2$ for vectors (A4). Page 8 then writes the kinetic term with radial coordinates, $(r_{j+1} - r_j)^2$. The difference of radii is not the distance between beads: $|\mathbf{r}_{j+1} - \mathbf{r}_j| \geq ||\mathbf{r}_{j+1}| - |\mathbf{r}_j||$, with equality only when the two beads lie on the same ray from the origin. So that expression is not the three-dimensional primitive action. Meanwhile H5 integrates with the three-dimensional measure $4\pi r^2 dr$, so the notes combine a one-dimensional kinetic term with a three-dimensional potential integral.
- Minor: in the first line of p. 6 the free propagator is written $\langle x_{j+1}|e^{-\beta\hat{K}}|x_j\rangle$, missing the Δ; the next line uses $\Delta\beta$ correctly.
- Minor: the identity written as "$\prod_j e^j = \sum_j e^j$" should read $\prod_j e^{a_j} = e^{\sum_j a_j}$. The next line uses the correct identity.

What H3 gets right: the ordering, with V evaluated at the right-hand point x_j, matches T; and the per-slice prefactor $(\mu/2\pi\hbar^2\Delta\beta)^{1/2}$ is the correct one-dimensional free kernel.

H4 (p. 8) is correct. With $V = -k/r$, $-\Delta\beta V = \Delta\beta k/r > 0$, and the weight tends to $+\infty$ as $r \to 0^+$; the notes correctly note that it is dimensionless. On its own, though, it does not imply divergence: unbounded functions such as $1/r$ are integrable in three dimensions. It needs H5.

H5 (p. 8) is correct but unproven.

- "r^2 does not compensate $e^{a/r}$" is an assertion. The proof is (B1) to (B3): $e^x \geq x^3/6$ gives $r^2 e^{a/r} \geq a^3/(6r)$, whose integral diverges logarithmically. The true growth with a cutoff is much faster than logarithmic: $I \simeq 4\pi\eta^4 e^{a/\eta}/a$ (B6).
- H5 concerns one bead with the kinetic factor ignored, so by itself it is not yet a statement about Z_m.
- On p. 9, item 1 reads "$e^{a/r} \to 0$". Taken literally this is false: the limit as $r \to 0^+$ is $+\infty$. I read it as shorthand for "$e^{a/r}$ as $r \to 0$", but I flag the reading as uncertain.

H6 (p. 9) has a correct conclusion and no proof. The kinetic factors indeed do not compensate, but the notes give no argument.

- The missing step is a lower bound on the kinetic factors on a set of positive measure that contains the singular configurations: (B7) to (B9). The obvious bound on a Gaussian is an upper bound, and that points the wrong way for proving a divergence.
- B.5 sharpens the conclusion: the kinetic factors do not change the leading growth at all (B16).
- "The sum diverges" does not distinguish the two divergences of Z_N. The notes do not mention the large-volume one (B13), which is present even for $V = 0$ and is the same mechanism that makes H1 fail.

My hypotheses against the notes'. I use none of H1 to H6 as a premise. My (A3) is the corrected three-dimensional form of H3; the lemma (B1) to (B3) proves H4 and H5; and (B7) to (B9) prove H6's conclusion.

Hypotheses of the notes I consider unnecessary for (a) to (c):

- H1, the ground-state extraction;

- H2, the Taylor-series justification, which the Trotter theorem replaces;
- the Wick-rotation route of p. 7, since (A3) follows directly from the operator expression.

Hypotheses I need that the notes do not state:

- three dimensions, with the measure d^3r;
- positivity of the integrand and Tonelli's theorem [A1];
- $V \leq 0$, so that every potential factor is at least 1;
- a lower bound on the kinetic factors on bounded sets;
- the large-volume divergence and its separation from the origin (B.3, B.4);
- the fact that T^N is not an operator whose trace can be taken (B.6);
- for Part C: Kato's theorem, the Kato-class kernel theorem, the Coulomb data at the origin, and the semiboundedness hypothesis of the Trotter formula.

4. Verification

4.1 Self-check battery

1. Dimensions. In atomic units every displayed quantity is a pure number or a power of a_0. Restoring units:
   - the exponent $\Delta\beta k/r$ is dimensionless (k in J · m, $\Delta\beta$ in J^{-1}, r in m), so $a = \Delta\beta k$ is a length, η^4/a in (B6) is a volume, and I is a volume, as it must be;
   - each slice prefactor $(\mu/2\pi\hbar^2\Delta\beta)^{3/2}$ has units m^{-3} and is paired with d^3r, so Z_N is dimensionless, as a trace must be;
   - $K_\tau(\mathbf{0},\mathbf{0})$, $(2\pi\tau)^{-3/2}$, $1/(\pi n^3)$ and the continuum term of (C5) all have units a_0^{-3}.
2. Limiting cases.
   - $\Delta\tau \to 0$ at fixed N: the coefficient $2\pi\Delta\tau^3/3$ in (B2) tends to 0, yet $I = \infty$ for every $\Delta\tau > 0$; the singular shell has thickness $\eta^2/\Delta\tau$.
   - $N \to \infty$ at fixed β: $Z_N = Z_N^{(L)} = \infty$ for every N, so there is no limit to take.
   - $\Delta\tau \to \infty$ or $\beta \to \infty$: still infinite.
   - Coupling: with $V = -Z/r$, every $Z > 0$ gives the same divergence; $Z = 0$ leaves only the large-volume divergence.
   - $\eta \to 0$: (B6) and (B16). $L \to \infty$: (B13).
3. Reduction to simpler cases.
   - $V = 0$: (A3) becomes $\operatorname{Tr} e^{-\beta K} = \infty$, by volume alone, and its localised version is finite, consistent with (B11) as $\delta \to 0$.
   - $N = 1$: $Z_1 = (2\pi\beta)^{-3/2} \int d^3r\, e^{\beta/r}$ is infinite at both ends, consistent with (B9) and (B13).
   - Repulsive $V = +1/r$: the integrand is bounded by its kinetic part and the localised trace is finite. The proof of (B9) breaks exactly at the step $w \geq 1$, where it should.
4. Symmetries. Rotations: Z_N depends only on the radii and relative angles (used in (B17)). Cyclic relabelling of the beads (used to single out bead 1). Reversing the bead order, or reversing the factor order in T, leaves (A3) unchanged (Part A).
5. Signs. The divergence comes from $-\Delta\tau V = +\Delta\tau/r > 0$ for the attractive potential, as in the problem statement ($V = -1/r$). The Euclidean kinetic exponent is negative.
6. Order of magnitude.
   - For $\Delta\tau = 0.1$ (for instance $\beta = 10$, $N = 100$) and $\eta = 10^{-3}a_0$, (B6) gives $I \approx 4\pi \cdot 10^{-12}e^{100}/0.1 \times 1.05 \approx 3.5 \times 10^{33}\, a_0^3$, against $|B_1| \approx 4.2\, a_0^3$.

- Even a finite proton, of radius about $1.6 \times 10^{-5} a_0$ [R], would leave a weight of order $e^{\Delta\tau/R_p} \approx e^{6250}$ at its centre. The primitive weight is unusable long before the mathematical divergence.
- For the exact kernel, $K_1(\mathbf{0},\mathbf{0})/(2\pi)^{-3/2} \approx 14.5$ [N, C4].

7. Boundary conditions. The trace imposes $\mathbf{r}_{N+1} = \mathbf{r}_1$ (A3); for $N = 1$ the single link has zero length. In C4, route (ii) imposes $u(0) = 0$ (the regular $l = 0$ solution) and $u(R_{\text{box}}) = 0$; changing the box size has no effect at the 10^{-13} level (G5).
8. Degenerate cases. Only nonnegative integrands are integrated, so the only exchanges of limit and integral are Tonelli and monotone convergence (B3). κ in (B17) never vanishes, because $r \geq \eta > 0$. $\Delta\tau > 0$ and $N \geq 1$ throughout. The orthogonality in (C7) comes from disjoint supports.

4.2 Computational checks

All scripts are standalone and run from a clean interpreter with `python3 -I`. Environment: Python 3.13.16, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0, sympy 1.14.0, Linux sandbox. In round 2 every script was re-run once, with code byte-identical to round 1, and the outputs reproduced below are from that round-2 run. Each round-2 output is byte-identical to its round-1 counterpart (compared with `cmp`). In round 1 each script had been run twice, also with identical results. Check C6, added in round 2 for finding F1.1, is in §10. C5 was run with two different quadrature methods, and both runs are reported. The only edit to any output is in C5 run 1: the absolute sandbox path in front of the script name in scipy's warning lines has been removed.

The equation numbers cited in the code comments come from an earlier draft of this document. The code is reproduced as it was run, so those comments are left unchanged. They map to the final numbering as follows:

Check	Number in the code comment	Final number
C1	(B4)–(B6)	(B2) and (B6)
C2	(B12)	(B17)
C2	(B13)–(B15)	(B16)–(B18)
C3	(B9)–(B10)	(B12)–(B13)
C4	(C9)	(C5)
C5	(C3)–(C5)	(C1)–(C3)

C1 [N]. One-bead radial integral against the growth law (B2) and (B6)

What it tests. It computes $4\pi \int_\eta^\varepsilon r^2 e^{a/r} dr$ by direct quadrature in r. The integrand comes straight from the potential factor $e^{-\Delta\tau V}$ with $V = -1/r$ and $a = \Delta\tau$, not from the derivation. The result is compared with the lower bound (B2), with the leading law and its first correction, and with the three-term law (B6). The failure criteria were declared in the script before it was run.

What it would have caught. A wrong power of η or a wrong exponent in the growth law; a wrong coefficient (for example $1/a$); a violation of the logarithmic lower bound. The contrast column, the ratio to the wrong law $\eta^3 e^{a/\eta}$, decays like η, which shows the check can tell neighbouring powers apart.

What it would not have caught. Errors in going from Z_N to the one-bead integral; the prefactor $(2\pi\Delta\tau)^{-3N/2}$; anything about the kinetic factor or the large-volume divergence. Rows with $a/\eta \leq 10$ are outside the asymptotic regime: they are reported and tested only against the lower bound.

Assumptions. The quadrature breakpoints are placed at multiples of η^2/a, a scale that comes from my analysis. A wrong scale would make the quadrature less accurate and show up as failures, not as passes. The mpmath working precision is 50 digits.

Outcome. All criteria pass. At $a/\eta = 160$: $I/L_1 = 1.0258$; $(I/L_1 - 1)/(\eta/a) = 4.130$, against a prediction of $4 + 20\eta/a = 4.125$; and $I/L_3 - 1 = 2.99 \times 10^{-5}$, against $120(\eta/a)^3 = 2.93 \times 10^{-5}$.

```
# C1: single-bead radial integral with an inner cutoff, against the predicted growth law.
# Environment: Python 3.13.16, mpmath 1.3.0. Deterministic (no random numbers).
#
# Quantity (from the bead integral, potential factor exp(-dtau*V) with V = -1/r, a = dtau):
#     I(eta) = 4*pi * int_eta^eps r^2 exp(a/r) dr          (integrated in r, directly)
# Predictions under test (derived in the answer, eqs. (B4)-(B6)):
#     lower bound      I(eta) >= (2*pi*a^3/3) * ln(eps/eta)          for all 0 < eta < eps
#     leading law      I(eta) ~ 4*pi*eta^4*exp(a/eta)/a              as eta -> 0
#     first correction (I/L1 - 1)/(eta/a) -> 4
#     three-term law   I/L3 - 1 = O((eta/a)^3), L3 = 4*pi*a^3*e^{a/eta}[(eta/a)^4 + 4(eta/a)^5
    + 20(eta/a)^6]
# Declared failure (before running):
#     F1 any lower-bound violation;
#     F2 |I/L1 - 1| > 0.05 at the largest a/eta (=160) for any a;
#     F3 (I/L1 - 1)/(eta/a) outside [3.6, 4.4] at a/eta = 160;
#     F4 |I/L3 - 1| > 200*(eta/a)^3 at a/eta >= 40 (the next term predicts ~120*(eta/a)^3).
# Contrast: the ratio to a wrong law eta^3*exp(a/eta) is printed to show the check can tell
    powers apart.
import mpmath as mp

mp.mp.dps = 50
eps = mp.mpf(1)
fails = []

def I_direct(a, eta):
    a = mp.mpf(a); eta = mp.mpf(eta)
    w = eta**2 / a  # width of the region where exp(a/r) is within a factor e of its maximum
    pts = [eta] + [eta + k * w for k in (1, 3, 10, 30, 100, 300, 1000) if eta + k * w < eps] +
    [eps]
    return 4 * mp.pi * mp.quad(lambda r: r**2 * mp.exp(a / r), pts)

print("a     a/eta   eta            I(eta)            I/L1          (I/L1-1)/(eta/a)   I/L3-1
            bound_ok  I/(eta^3 e^{a/eta})")
for a in (mp.mpf('0.5'), mp.mpf(1), mp.mpf(2)):
    for q in (2, 5, 10, 20, 40, 80, 160):
        eta = a / q
        if eta >= eps:
            continue
        I = I_direct(a, eta)
        x = eta / a
        L1 = 4 * mp.pi * eta**4 * mp.exp(a / eta) / a
        L3 = 4 * mp.pi * a**3 * mp.exp(a / eta) * (x**4 + 4 * x**5 + 20 * x**6)
        lb = (2 * mp.pi * a**3 / 3) * mp.log(eps / eta)
        ok = I >= lb
        if not ok:
            fails.append(("F1", float(a), q))
        r1 = I / L1
        c1 = (r1 - 1) / x
        r3 = I / L3 - 1
        wrong = I / (eta**3 * mp.exp(a / eta))
        print(f"{float(a):<5} {q:<7} {mp.nstr(eta,8):<14} {mp.nstr(I,10):<17} {mp.nstr(r1,10)
    :<13} {mp.nstr(c1,8):<18} {mp.nstr(r3,6):<15} {str(ok):<9} {mp.nstr(wrong,6)}")
        if q == 160:
            if abs(r1 - 1) > 0.05:
                fails.append(("F2", float(a), q))
            if not (3.6 <= c1 <= 4.4):
                fails.append(("F3", float(a), q))
```

```
        if q >= 40 and abs(r3) > 200 * x**3:
            fails.append(("F4", float(a), q))

print("FAILURES:", fails if fails else "none")
```

```
a     a/eta   eta            I(eta)            I/L1          (I/L1-1)/(eta/a)   I/L3-1
     bound_ok  I/(eta^3 e^{a/eta})
0.5   2       0.25           8.73960349        12.04766468   22.095329          0.505958
     True      75.6977
0.5   5       0.1            9.950773425       26.67746559   128.38733          9.26056
     True      67.0478
0.5   10      0.05           15.9913494        4.621898632   36.218986          1.88869
     True      5.80805
0.5   20      0.025          6080.139943       1.276509475   5.5301895          0.0212076
     True      0.802055
0.5   40      0.0125         1.610095881e+11   1.114788683   4.5915473          0.00205724
     True      0.350221
0.5   80      0.00625        2.238228677e+27   1.053382198   4.2705758          0.000244223
     True      0.165465
0.5   160     0.003125       7.547876444e+60   1.025811897   4.1299034          2.98763e-5
     True      0.0805671
1.0   2       0.5            13.70402039       2.361397781   2.7227956          -0.704825
     True      14.8371
1.0   5       0.2            23.39337986       7.839542463   34.197712          2.01521
     True      19.7029
1.0   10      0.1            71.71798763       2.59103579    15.910358          0.619397
     True      3.25599
1.0   20      0.05           48584.90674       1.275034259   5.5006852          0.0200274
     True      0.801128
1.0   40      0.025          1.288076705e+12   1.114788683   4.5915473          0.00205724
     True      0.350221
1.0   80      0.0125         1.790582942e+28   1.053382198   4.2705758          0.000244223
     True      0.165465
1.0   160     0.00625        6.038301155e+61   1.025811897   4.1299034          2.98763e-5
     True      0.0805671
2.0   5       0.4            77.51487582       3.247078682   11.235393          0.248876
     True      8.1608
2.0   10      0.2            464.111738        2.095935349   10.959353          0.30996
     True      2.63383
2.0   20      0.1            388569.6217       1.274674618   5.4934924          0.0197397
     True      0.800902
2.0   40      0.05           1.030461364e+13   1.114788683   4.5915473          0.00205724
     True      0.350221
2.0   80      0.025          1.432466353e+29   1.053382198   4.2705758          0.000244223
     True      0.165465
2.0   160     0.0125         4.830640924e+62   1.025811897   4.1299034          2.98763e-5
     True      0.0805671
FAILURES: none
```

C2 [N]. Can the kinetic factors compensate? The $N = 2$ integral with a cutoff, (B16) to (B18)

What it tests. In part (a), it uses Monte Carlo over independent random directions to test the angular identity (B17) independently of its derivation (seed 20261019, 400 000 pairs per κ). In part (b), it computes $R(\eta)$, the full $N = 2$ bead integral with cutoff divided by the same integral with the kinetic factor replaced by its maximum, for $\Delta\tau = 1$, $L = 1.5$, and five values of η. The input is the $N = 2$ integrand of (A3), which comes from the problem statement and the given kernel. "The kinetic factor compensates" would show up as $R \to 0$.

What it would have caught. Compensation; a wrong kinetic exponent at $N = 2$; a wrong angular reduction (through part (a)); a wrong small-η expansion (B18).

What it would not have caught. Anything at $N \geq 3$, which only the general bounds (B14) to (B16) cover; the large-volume divergence, since the beads are confined to B_L; the overall prefactor, which cancels in R.

Assumptions. [A] Two independent, uniformly distributed directions have $\cos\theta_{12}$ uniform on $[-1, 1]$; this is what (B17) uses. The Monte Carlo samples directions directly as normalised Gaussian vectors, so it does not rely on that fact.

Outcome. All criteria pass. R increases towards 1, and at $\eta = 0.0125$, $(1 - R)/(2\eta^2/\Delta\tau) = 1.0272$, against a prediction of about $1 + 2\eta/\Delta\tau = 1.025$. At $\eta = 0.2$ and 0.1 the region away from the origin still dominates and R is small; this is expected, and no criterion was placed on those rows except $R \leq 1$ and the monotonic decrease of $1 - R$.

```
# C2: do the kinetic factors compensate? N = 2 primitive bead integral with an inner cutoff eta
     on both
# beads and both beads confined to |r| < L.
# Environment: Python 3.13.16, numpy 2.5.3, mpmath 1.3.0. Seed fixed below.
#
# From the bead integral (answer eq. (A3)) at N = 2, the two links give |r2-r1|^2/(2 dtau)
    twice, so
#   Z2(eta) = (2 pi dtau)^(-3) * int_{eta<|r1|,|r2|<L} d^3r1 d^3r2 exp(-|r1-r2|^2/dtau + dtau/
    r1 + dtau/r2).
# Angular integration (answer eq. (B12)), valid for independent uniform directions:
#   int dOmega1 dOmega2 exp(-|r1-r2|^2/dtau) = 16 pi^2 exp(-(r1^2+r2^2)/dtau) sinh(k)/k,  k = 2
     r1 r2/dtau.
# Part (a) tests that angular identity by Monte Carlo over random directions (independent of
    the closed form).
# Part (b) computes R(eta) = Z2(eta) / [ (2 pi dtau)^(-3) I(eta)^2 ],  I(eta) = 4 pi int_eta^L
    r^2 e^{dtau/r} dr,
#   i.e. the full N = 2 integral divided by what it would be if the kinetic factor were
    replaced by its maximum.
# Predictions (answer eqs. (B13)-(B15)):  R <= 1 for every eta;  R -> 1 as eta -> 0;
#   (1 - R)/(2 eta^2/dtau) -> 1, with a first correction of about +2 eta/dtau.
# Declared failure (before running):
#   Fa  any Monte Carlo deviation larger than 5 standard errors;
#   Fb1 R > 1 + 1e-12 for any eta;
#   Fb2 1 - R not decreasing as eta decreases;
#   Fb3 (1 - R)/(2 eta^2/dtau) outside [0.95, 1.10] at the smallest eta (prediction ~1.025).
# What would count as "the kinetic factor compensates": R -> 0 as eta -> 0.
import numpy as np
import mpmath as mp

rng = np.random.default_rng(20261019)
fails = []

# ---------- (a) Monte Carlo test of the angular identity ----------
def unit_vectors(n):
    v = rng.normal(size=(n, 3))
    return v / np.linalg.norm(v, axis=1, keepdims=True)

n = 400_000
print("(a) angular average <exp(k cos theta)> over independent uniform directions vs sinh(k)/k"
    )
for k in (0.01, 0.5, 2.0, 5.0):
    u, w = unit_vectors(n), unit_vectors(n)
    vals = np.exp(k * np.sum(u * w, axis=1))
    est, se = vals.mean(), vals.std(ddof=1) / np.sqrt(n)
    exact = np.sinh(k) / k
    z = (est - exact) / se
    print(f"  k={k:<5} MC={est:.6f} +- {se:.2e}   sinh(k)/k={exact:.6f}   z={z:+.2f}")
    if abs(z) > 5:
        fails.append(("Fa", k))

# ---------- (b) the N = 2 integral ----------
```

```
mp.mp.dps = 30
dtau, L = mp.mpf(1), mp.mpf('1.5')

def pts(eta):
    w = eta**2 / dtau
    return [eta] + [eta + k * w for k in (1, 3, 10, 30, 100, 300, 1000) if eta + k * w < L] + [
    L]

def g(r1, r2):  # angular average of the kinetic factor, divided by 16 pi^2
    k = 2 * r1 * r2 / dtau
    return mp.exp(-(r1**2 + r2**2) / dtau) * mp.sinh(k) / k

print("\n(b) R(eta) = Z2 / [(2 pi dtau)^-3 I^2],  dtau = 1, L = 1.5")
print("  eta        R                    1-R                (1-R)/(2 eta^2/dtau)   predicted
    ~1+2eta/dtau")
prev = None
for eta in (mp.mpf('0.2'), mp.mpf('0.1'), mp.mpf('0.05'), mp.mpf('0.025'), mp.mpf('0.0125')):
    P = pts(eta)
    wgt = lambda r: r**2 * mp.exp(dtau / r - dtau / eta)          # scaled radial weight
    Iw = mp.quad(wgt, P)                                            # I(eta) / (4 pi e^{dtau/
    eta})
    J = mp.quad(lambda a, b: wgt(a) * wgt(b) * g(a, b), P, P)       # Z2 / [(2 pi dtau)^-3 16
    pi^2 e^{2 dtau/eta}]
    one_minus = mp.quad(lambda a, b: wgt(a) * wgt(b) * (1 - g(a, b)), P, P) / Iw**2
    R = J / Iw**2
    ratio = one_minus / (2 * eta**2 / dtau)
    print(f"  {mp.nstr(eta,5):<10} {mp.nstr(R,16):<20} {mp.nstr(one_minus,10):<19} {mp.nstr(
    ratio,8):<22} {mp.nstr(1+2*eta/dtau,6)}")
    print(f"             consistency |R + (1-R) - 1| = {mp.nstr(abs(R + one_minus - 1), 3)}")
    if R > 1 + mp.mpf('1e-12'):
        fails.append(("Fb1", float(eta)))
    if prev is not None and not (one_minus < prev):
        fails.append(("Fb2", float(eta)))
    prev = one_minus
    last_ratio = ratio
if not (0.95 <= last_ratio <= 1.10):
    fails.append(("Fb3", float(last_ratio)))
print("FAILURES:", fails if fails else "none")
```

```
(a) angular average <exp(k cos theta)> over independent uniform directions vs sinh(k)/k
  k=0.01  MC=1.000015 +- 9.13e-06   sinh(k)/k=1.000017   z=-0.19
  k=0.5   MC=1.042037 +- 4.71e-04   sinh(k)/k=1.042191   z=-0.33
  k=2.0   MC=1.811437 +- 2.97e-03   sinh(k)/k=1.813430   z=-0.67
  k=5.0   MC=14.880139 +- 4.71e-02   sinh(k)/k=14.840642   z=+0.84

(b) R(eta) = Z2 / [(2 pi dtau)^-3 I^2],  dtau = 1, L = 1.5
  eta        R                    1-R                (1-R)/(2 eta^2/dtau)   predicted ~1+2eta/
    dtau
  0.2        0.2747484332427588   0.7252515668       9.0656446              1.4
             consistency |R + (1-R) - 1| = 1.97e-31
  0.1        0.549999669688688    0.4500003303       22.500017              1.2
             consistency |R + (1-R) - 1| = 3.71e-29
  0.05       0.9931740416191257   0.006825958381     1.3651917              1.1
             consistency |R + (1-R) - 1| = 6.9e-25
  0.025      0.998674583540708    0.001325416459     1.0603332              1.05
             consistency |R + (1-R) - 1| = 3.43e-23
  0.0125     0.9996789889021237   0.0003210110979    1.0272355              1.025
             consistency |R + (1-R) - 1| = 6.5e-22
FAILURES: none
```

C3 [N]. Composition law of the free kernel, (B12)

What it tests. (i) The completing-the-square identity of B.4, symbolically, with the two sides written independently. (ii) The one-dimensional composition integral by quadrature, against the closed form, at twelve parameter sets, including extreme ratios s/t and large separations.

What it would have caught. An error in the composition law or in its normalisation, either of which would invalidate (B13).

What it would not have caught. Anything involving V.

Assumptions. [A] $s, t > 0$, declared in the code; imaginary-time steps are positive.

Outcome. All twelve parameter sets agree to 1.2×10^{-15} or better, and the symbolic difference is identically zero.

```
# C3: composition law of the free kernel, used in the large-r (volume) argument, eq. (B9)-(B10)
    .
# Environment: Python 3.13.16, numpy 2.5.3, scipy 1.18.1, sympy 1.14.0. Seed fixed below.
#
# Claim (1D factor; the 3D kernel is the product of three such factors):
#   int dy rho1(x,y;s) rho1(y,z;t) = rho1(x,z;s+t),   rho1(x,y;t) = (2 pi t)^(-1/2) exp(-(x-y)
    ^2/(2t)).
# (i) Symbolic: the completing-the-square identity of eq. (B9), written out independently on
    both sides.
# (ii) Numerical: quadrature of the left side vs the right side at random (x,z,s,t), including
    points near
#      the edges (s/t very small or very large, |x-z| several widths). Tolerance 1e-9 relative.
# Declared failure: symbolic difference not identically zero, or any relative discrepancy > 1e
    -9.
import numpy as np
import sympy as sp
from scipy.integrate import quad

x, y, z = sp.symbols('x y z', real=True)
s, t = sp.symbols('s t', positive=True)   # [A] s, t > 0: imaginary-time steps are positive
lhs = -(x - y)**2 / (2 * s) - (y - z)**2 / (2 * t)
rhs = -(s + t) / (2 * s * t) * (y - (t * x + s * z) / (s + t))**2 - (x - z)**2 / (2 * (s + t))
print("(i) symbolic: expand(lhs - rhs) =", sp.simplify(sp.expand(lhs - rhs)))
fails = []
if sp.simplify(sp.expand(lhs - rhs)) != 0:
    fails.append("symbolic")

rng = np.random.default_rng(7)
rho = lambda a, b, tt: np.exp(-(a - b) ** 2 / (2 * tt)) / np.sqrt(2 * np.pi * tt)
cases = [(rng.uniform(-3, 3), rng.uniform(-3, 3), rng.uniform(0.05, 3), rng.uniform(0.05, 3))
    for _ in range(8)]
cases += [(0.0, 0.0, 1e-3, 5.0), (2.0, -2.0, 5.0, 1e-3), (0.0, 6.0, 0.5, 0.5), (1.0, 1.0, 0.01,
     0.01)]
print("(ii) numerical composition, x z s t -> relative error")
for (xx, zz, ss, tt) in cases:
    c = (tt * xx + ss * zz) / (ss + tt); w = np.sqrt(ss * tt / (ss + tt))
    val, err = quad(lambda yy: rho(xx, yy, ss) * rho(yy, zz, tt), c - 40 * w, c + 40 * w,
    epsabs=0, epsrel=1e-13, limit=200)
    ref = rho(xx, zz, ss + tt)
    rel = abs(val - ref) / ref
    print(f"  {xx:+.3f} {zz:+.3f} {ss:.4f} {tt:.4f}  ->  {rel:.2e}")
    if rel > 1e-9:
        fails.append((xx, zz, ss, tt))
print("FAILURES:", fails if fails else "none")
```

```
(i) symbolic: expand(lhs - rhs) = 0
(ii) numerical composition, x z s t -> relative error
  +0.751 +2.383 2.3383 0.7144  ->  0.00e+00
  -1.199 +2.241 0.0655 2.4726  ->  1.43e-16
  +1.782 -0.192 0.9439 0.8714  ->  0.00e+00
```

```
  -1.471 -0.330 1.5384 1.6828  ->  0.00e+00
  +2.973 +1.756 1.8854 2.9674  ->  1.79e-16
  -1.708 -2.039 1.8570 0.1796  ->  2.04e-16
  -2.786 +0.089 1.4253 2.7556  ->  1.91e-16
  +0.775 +0.085 1.5158 0.7802  ->  0.00e+00
  +0.000 +0.000 0.0010 5.0000  ->  1.56e-16
  +2.000 -2.000 5.0000 0.0010  ->  1.16e-15
  +0.000 +6.000 0.5000 0.5000  ->  4.08e-16
  +1.000 +1.000 0.0100 0.0100  ->  1.57e-16
FAILURES: none
```

C4 [N]. The exact kernel at the origin, by two independent routes, (C5)

What it tests. Route (i) evaluates the closed form (C5), whose inputs are recalled spectral data. Route (ii) uses only the problem's Hamiltonian: it discretises the $l = 0$ radial operator $-\frac{1}{2}u'' - u/r$ on $(0, R_{\text{box}})$ by second-order finite differences, diagonalises it, and assembles $\frac{1}{4\pi}\sum_i e^{-\tau E_i} u_i'(0)^2$. A control run with $V = 0$ must reproduce $(2\pi\tau)^{-3/2}$. Three grid spacings are used, plus a box-size test, and the lowest three box eigenstates are compared with the recalled bound-state data.

What it would have caught.

- A wrong Sommerfeld factor. The repulsive form, for example, gives a continuum term orders of magnitude too small at small k.
- A wrong bound-state density at the origin.
- A missing factor 4π or $(2\pi)^3$.
- A wrong finite-difference normalisation (control G1).
- A kernel that diverged at the origin: route (ii) would then fail to converge in h.
- A box-size artifact (G5).

What it would not have caught. Finiteness at points other than the origin; off-diagonal elements; the general theorem cited in C.3. Both routes also assume that the $l = 0$ sector carries all the weight at $r = 0$. That holds because $R_{nl}(0) = 0$ for $l \geq 1$, and the same is true of the continuum partial waves [R].

Assumptions.

- In route (ii), the $l = 0$ sector of the self-adjoint H corresponds to the boundary condition $u(0) = 0$. The radial operator is limit-circle at $r = 0$, and $u(0) = 0$ selects the regular solution, which is the one corresponding to H^2 functions [R]. Agreement with route (i), whose inputs are the spectral data of the three-dimensional H, supports this. [A]
- $u'(0)$ is taken from the one-sided second-order formula $(4u_1 - u_2)/2h$, which is valid for u smooth at 0 (for example $u_{1s} \propto re^{-r}$). [A]
- Eigenpairs above $E = 120$ are dropped. The bound on the dropped part, below 10^{-19}, is stated in the script.

Outcome. All criteria pass.

- The observed convergence order is 1.96 to 1.99.
- After Richardson extrapolation, routes (i) and (ii) agree to 5×10^{-7} relative or better at all four τ.
- The free control converges at second order; at the finest grid its error is between 1.6×10^{-5} ($\tau = 4$) and 1.3×10^{-4} ($\tau = 0.5$).
- The box bound states reproduce $E_n = -1/(2n^2)$ to 3×10^{-6}, and $1/(\pi n^3)$ to 3×10^{-4}, for $n = 1, 2, 3$.

```
# C4: the exact kernel <0|exp(-tau H)|0> for H = -(1/2) Laplacian - 1/r, by two independent
    routes.
# Environment: Python 3.13.16, numpy 2.5.3, scipy 1.18.1, mpmath 1.3.0. Deterministic.
#
# Route (i), closed form from recalled spectral data, answer eq. (C9):
#   K0(tau) = sum_{n>=1} exp(tau/(2n^2)) / (pi n^3)  +  (1/pi) int_0^inf k exp(-tau k^2/2) / (1
     - exp(-2 pi/k)) dk
#   inputs [R]/[V]: |psi_n00(0)|^2 = 1/(pi n^3), E_n = -1/(2n^2), Sommerfeld factor (2pi/k)/(1-
    e^{-2pi/k}),
#   completeness of {bound states, Coulomb waves}, and that only l = 0 states are nonzero at r
    = 0.
# Route (ii), from the Hamiltonian only: the l = 0 radial operator -(1/2) d^2/dr^2 - 1/r on (0,
     Rbox) with
#   u(0) = u(Rbox) = 0, discretised by second-order finite differences, diagonalised. For l =
    0,
#   psi(r) = u(r)/(sqrt(4 pi) r), so psi(0) = u'(0)/sqrt(4 pi); states with l >= 1 vanish at r
    = 0.
#   K0_box = (1/4pi) sum_i exp(-tau E_i) u_i'(0)^2, with u'(0) from the second-order one-sided
    formula
#   u'(0) ~ (4 u_1 - u_2)/(2h). Eigenpairs above Ecut = 120 are dropped; the dropped part is
    bounded by
#   exp(-tau_min*Ecut) * 17/(16 pi h^3) < 1e-19 (completeness of the discrete eigenvectors).
# Control: the same box code with V = 0 must reproduce the free value (2 pi tau)^(-3/2).
# Declared failure (before running):
#   G1 free control: relative error > 1e-3 at the finest h;
#   G2 Coulomb box vs closed form: relative difference > 5e-3 at the finest h;
#   G3 after Richardson extrapolation with the observed order: relative difference > 5e-4;
#   G4 box bound states n = 1..3: |E_n + 1/(2n^2)| > 1e-3 or relative error of |psi(0)|^2 vs
    1/(pi n^3) > 5e-3;
#   G5 box-size sensitivity (Rbox = 30 vs 40 at h = 0.01): relative change > 1e-6.
import numpy as np
from scipy.linalg import eigh_tridiagonal
import mpmath as mp

taus = [0.5, 1.0, 2.0, 4.0]
ECUT = 120.0
fails = []

# ---------------- route (i) ----------------
mp.mp.dps = 30
def closed_form(tau):
    tau = mp.mpf(tau)
    Sb = mp.nsum(lambda n: mp.exp(tau / (2 * n**2)) / (mp.pi * n**3), [1, mp.inf])
    Sc = mp.quad(lambda k: k * mp.exp(-tau * k**2 / 2) / (-mp.expm1(-2 * mp.pi / k)), [0, 0.5,
    2, 6, mp.inf]) / mp.pi
    return float(Sb), float(Sc), float(Sb + Sc)

# ---------------- route (ii) ----------------
def box(h, Rbox, coulomb=True):
    M = int(round(Rbox / h)) - 1
    r = h * np.arange(1, M + 1)
    d = np.full(M, 1.0 / h**2) + (-1.0 / r if coulomb else 0.0)
    e = np.full(M - 1, -0.5 / h**2)
    w, v = eigh_tridiagonal(d, e, select='v', select_range=(-10.0, ECUT))
    du0 = (4 * v[0, :] - v[1, :]) / (2 * h**1.5)   # u'(0) with u_j = v_j / sqrt(h)
    return w, du0

def K_box(w, du0, tau):
    return float(np.sum(np.exp(-tau * w) * du0**2) / (4 * np.pi))

print("route (i): closed form")
cf = {}
```

```
for tau in taus:
    Sb, Sc, tot = closed_form(tau)
    cf[tau] = tot
    free = (2 * np.pi * tau) ** -1.5
    print(f"  tau={tau:<4} bound part={Sb:.10f}  continuum part={Sc:.10f}  total={tot:.10f}
    free={free:.10f}  ratio={tot/free:.4f}")

hs = [0.02, 0.01, 0.005]
Rbox = 40.0
res, resfree = {}, {}
for h in hs:
    w, du0 = box(h, Rbox, True)
    wf, du0f = box(h, Rbox, False)
    res[h] = {tau: K_box(w, du0, tau) for tau in taus}
    resfree[h] = {tau: K_box(wf, du0f, tau) for tau in taus}
    if h == hs[-1]:
        print("\nG4: lowest box eigenstates at h = 0.005, Rbox = 40")
        for n in (1, 2, 3):
            En, p0 = w[n - 1], du0[n - 1] ** 2 / (4 * np.pi)
            dE, dp = En + 1 / (2 * n**2), p0 / (1 / (np.pi * n**3)) - 1
            print(f"  n={n}: E={En:.7f} (exact -1/(2n^2) = {-1/(2*n**2):.7f}, diff {dE:+.2e});
     |psi(0)|^2={p0:.7f} (1/(pi n^3) = {1/(np.pi*n**3):.7f}, rel {dp:+.2e})")
            if abs(dE) > 1e-3 or abs(dp) > 5e-3:
                fails.append(("G4", n))

print("\nG1: free control, box (V = 0) vs (2 pi tau)^(-3/2)")
for tau in taus:
    free = (2 * np.pi * tau) ** -1.5
    row = "  ".join(f"h={h}: {resfree[h][tau]/free-1:+.2e}" for h in hs)
    print(f"  tau={tau:<4} relative error  {row}")
    if abs(resfree[hs[-1]][tau] / free - 1) > 1e-3:
        fails.append(("G1", tau))

print("\nG2/G3: Coulomb box vs closed form")
for tau in taus:
    d = [res[h][tau] / cf[tau] - 1 for h in hs]
    p = np.log2(abs(res[hs[0]][tau] - res[hs[1]][tau]) / abs(res[hs[1]][tau] - res[hs[2]][tau])
    )
    rich = res[hs[2]][tau] + (res[hs[2]][tau] - res[hs[1]][tau]) / (2**p - 1)
    dr = rich / cf[tau] - 1
    print(f"  tau={tau:<4} rel.diff h=0.02: {d[0]:+.2e}  h=0.01: {d[1]:+.2e}  h=0.005: {d
    [2]:+.2e}   observed order p={p:.2f}   Richardson: {dr:+.2e}")
    if abs(d[2]) > 5e-3:
        fails.append(("G2", tau))
    if abs(dr) > 5e-4:
        fails.append(("G3", tau))

print("\nG5: box size, h = 0.01")
w30, d30 = box(0.01, 30.0, True)
for tau in taus:
    a, b = K_box(w30, d30, tau), res[0.01][tau]
    print(f"  tau={tau:<4} Rbox=30: {a:.12f}  Rbox=40: {b:.12f}  rel.change {a/b-1:+.2e}")
    if abs(a / b - 1) > 1e-6:
        fails.append(("G5", tau))
print("\nFAILURES:", fails if fails else "none")
```

```
route (i): closed form
  tau=0.5  bound part=0.4760580540  continuum part=0.6693036422  total=1.1453616962  free
    =0.1795871221  ratio=6.3777
  tau=1.0  bound part=0.5953441704  continuum part=0.3242252408  total=0.9195694112  free
    =0.0634936359  ratio=14.4829
  tau=2.0  bound part=0.9427694300  continuum part=0.1599918269  total=1.1027612570  free
    =0.0224483903  ratio=49.1243
```

```
  tau=4.0  bound part=2.4461191758  continuum part=0.0796630394  total=2.5257822153  free
    =0.0079367045  ratio=318.2407

G4: lowest box eigenstates at h = 0.005, Rbox = 40
  n=1: E=-0.4999969 (exact -1/(2n^2) = -0.5000000, diff +3.12e-06);  |psi(0)|^2=0.3182901 (1/(
    pi n^3) = 0.3183099, rel -6.22e-05)
  n=2: E=-0.1249998 (exact -1/(2n^2) = -0.1250000, diff +1.95e-07);  |psi(0)|^2=0.0397874 (1/(
    pi n^3) = 0.0397887, rel -3.42e-05)
  n=3: E=-0.0555542 (exact -1/(2n^2) = -0.0555556, diff +1.36e-06);  |psi(0)|^2=0.0117929 (1/(
    pi n^3) = 0.0117893, rel +3.06e-04)

G1: free control, box (V = 0) vs (2 pi tau)^(-3/2)
  tau=0.5  relative error  h=0.02: +2.10e-03  h=0.01: +5.25e-04  h=0.005: +1.31e-04
  tau=1.0  relative error  h=0.02: +1.05e-03  h=0.01: +2.63e-04  h=0.005: +6.56e-05
  tau=2.0  relative error  h=0.02: +5.25e-04  h=0.01: +1.31e-04  h=0.005: +3.28e-05
  tau=4.0  relative error  h=0.02: +2.63e-04  h=0.01: +6.56e-05  h=0.005: +1.64e-05

G2/G3: Coulomb box vs closed form
  tau=0.5  rel.diff h=0.02: +3.19e-04  h=0.01: +8.16e-05  h=0.005: +2.07e-05   observed order p
    =1.96   Richardson: -4.64e-07
  tau=1.0  rel.diff h=0.02: -4.90e-04  h=0.01: -1.23e-04  h=0.005: -3.08e-05   observed order p
    =1.99   Richardson: +1.27e-07
  tau=2.0  rel.diff h=0.02: -8.87e-04  h=0.01: -2.23e-04  h=0.005: -5.60e-05   observed order p
    =1.99   Richardson: +3.69e-07
  tau=4.0  rel.diff h=0.02: -1.13e-03  h=0.01: -2.84e-04  h=0.005: -7.12e-05   observed order p
    =1.99   Richardson: +4.62e-07

G5: box size, h = 0.01
  tau=0.5  Rbox=30: 1.145455211031  Rbox=40: 1.145455211031  rel.change +5.48e-14
  tau=1.0  Rbox=30: 0.919456275654  Rbox=40: 0.919456275654  rel.change +1.21e-13
  tau=2.0  Rbox=30: 1.102515033371  Rbox=40: 1.102515033371  rel.change +1.55e-13
  tau=4.0  Rbox=30: 2.525065089752  Rbox=40: 2.525065089752  rel.change +2.23e-13

FAILURES: none
```

C5 [N]. The Kato-class quantity, (C1) to (C3)

What it tests. It computes $f_\delta(\mathbf{x})$ by quadrature in coordinates centred at $\mathbf{x}$, for $|\mathbf{x}|/\delta \in \{0, 0.1, 0.5, 0.99, 1, 1.5, 3\}$ and $\delta \in \{0.1, 1\}$. It compares the result with the derived bound $4\pi\delta$, and with a closed form based on Newton's theorem [R], which is not used in the derivation.

What it would have caught. A wrong bound (C3).

What it would not have caught. Whether the Kato class is the right hypothesis for the cited kernel theorem.

Run 1 failed. It used scipy's QUADPACK and reported disagreement with the recalled Newton formula at $|\mathbf{x}|/\delta = 0.99$ (1.6×10^{-7} against a tolerance of 10^{-7}). QUADPACK issued non-convergence warnings during the run. Python shows each warning only once, so the run does not say which rows triggered them. The inner integrand has an integrable singularity at $c = -1$ when $s = |\mathbf{x}|$, which lies inside the integration range for every $0 < |\mathbf{x}| < \delta$. The comment at the head of the run-2 script attributes the warnings to the 0.99 row specifically; that overstates what run 1 shows. The bound check passed at every point in run 1.

What I changed. I changed the check, not the derivation: only the quadrature method (to mpmath's tanh-sinh, with the outer integral split at $s = |\mathbf{x}|$). The tolerance, the formula and the bound are unchanged. Run 2 passes all criteria, with the bound attained at $\mathbf{x} = \mathbf{0}$.

Run 1 code and output:

```
# C5: the Kato-class quantity for V = -1/r in three dimensions, eq. (C3)-(C5).
# Environment: Python 3.13.16, numpy 2.5.3, scipy 1.18.1. Deterministic.
#
#   f_delta(x) = int_{|y-x|<delta} d^3y / (|y| |x-y|)
```

```
# Claim derived in the answer (AM-GM plus the bathtub comparison): f_delta(x) <= 4 pi delta for
     every x.
# Recalled comparison formula [R] (Newton's theorem for the shell average of 1/|x+s w|):
#   f_delta(x) = 4 pi (delta - |x|/2) for |x| <= delta,  2 pi delta^2/|x| for |x| >= delta.
# Computed here by direct quadrature in coordinates centred at x: y = x + s w, d^3y = s^2 ds
    dOmega,
#   f = 2 pi int_0^delta s ds int_{-1}^{1} dc (|x|^2 + s^2 + 2|x| s c)^(-1/2).
# Declared failure: any value exceeding 4 pi delta (1 + 1e-8), or disagreement with the
    recalled
# formula by more than 1e-7 relative (the latter would falsify only the [R] formula, not the
    bound).
import numpy as np
from scipy.integrate import quad

fails = []
print(" delta   |x|/delta   f (quadrature)     4 pi delta      Newton formula [R]   rel.diff")
for delta in (0.1, 1.0):
    for q in (0.0, 0.1, 0.5, 0.99, 1.0, 1.5, 3.0):
        X = q * delta
        def inner(s_):
            if X == 0.0:
                return 2.0 / s_ * s_  # int_{-1}^{1} dc / s = 2/s, times s
            g = lambda c: 1.0 / np.sqrt(X * X + s_ * s_ + 2 * X * s_ * c)
            pts = [-1.0, 1.0]
            val, _ = quad(g, -1.0, 1.0, epsabs=0, epsrel=1e-12, limit=200, points=None if abs(
    s_ - X) > 1e-12 else None)
            return s_ * val
        f, _ = quad(inner, 0.0, delta, epsabs=0, epsrel=1e-11, limit=400, points=[X] if 0 < X <
     delta else None)
        f *= 2 * np.pi
        newton = 4 * np.pi * (delta - X / 2) if X <= delta else 2 * np.pi * delta**2 / X
        rel = abs(f - newton) / newton
        print(f" {delta:<6} {q:<10}  {f:.12f}   {4*np.pi*delta:.12f}   {newton:.12f}    {rel:.1
    e}")
        if f > 4 * np.pi * delta * (1 + 1e-8):
            fails.append(("bound", delta, q))
        if rel > 1e-7:
            fails.append(("newton", delta, q))
print("FAILURES:", fails if fails else "none")
```

```
c5_kato_class_run1_scipy.py:25: IntegrationWarning: The algorithm does not converge.  Roundoff
    error is detected
  in the extrapolation table.  It is assumed that the requested tolerance
  cannot be achieved, and that the returned result (if full_output = 1) is
  the best which can be obtained.
  val, _ = quad(g, -1.0, 1.0, epsabs=0, epsrel=1e-12, limit=200, points=None if abs(s_ - X) > 1
    e-12 else None)
c5_kato_class_run1_scipy.py:27: IntegrationWarning: The maximum number of subdivisions (400)
    has been achieved.
  If increasing the limit yields no improvement it is advised to analyze
  the integrand in order to determine the difficulties.  If the position of a
  local difficulty can be determined (singularity, discontinuity) one will
  probably gain from splitting up the interval and calling the integrator
  on the subranges.  Perhaps a special-purpose integrator should be used.
  f, _ = quad(inner, 0.0, delta, epsabs=0, epsrel=1e-11, limit=400, points=[X] if 0 < X < delta
     else None)
 delta   |x|/delta   f (quadrature)     4 pi delta      Newton formula [R]   rel.diff
 0.1    0.0         1.256637061436   1.256637061436   1.256637061436    0.0e+00
 0.1    0.1         1.193805208364   1.256637061436   1.193805208364    0.0e+00
 0.1    0.5         0.942477796077   1.256637061436   0.942477796077    3.5e-16
 0.1    0.99        0.634601818606   1.256637061436   0.634601716025    1.6e-07
 0.1    1.0         0.628318530718   1.256637061436   0.628318530718    5.3e-16
 0.1    1.5         0.418879020479   1.256637061436   0.418879020479    0.0e+00
```

```
 0.1    3.0          0.209439510239   1.256637061436   0.209439510239    1.3e-16
 1.0    0.0          12.566370614359   12.566370614359   12.566370614359    0.0e+00
 1.0    0.1          11.938052083641   12.566370614359   11.938052083641    1.5e-16
 1.0    0.5          9.424777960769   12.566370614359   9.424777960769    0.0e+00
 1.0    0.99         6.346018199967   12.566370614359   6.346017160251    1.6e-07
 1.0    1.0          6.283185307180   12.566370614359   6.283185307180    0.0e+00
 1.0    1.5          4.188790204786   12.566370614359   4.188790204786    2.1e-16
 1.0    3.0          2.094395102393   12.566370614359   2.094395102393    2.1e-16
FAILURES: [('newton', 0.1, 0.99), ('newton', 1.0, 0.99)]
```

Run 2 code and output:

```
# C5 (run 2): the Kato-class quantity for V = -1/r in three dimensions, eq. (C3)-(C5).
# Environment: Python 3.13.16, mpmath 1.3.0. Deterministic.
# CHANGE FROM RUN 1 (c5_kato_class_run1_scipy.py): run 1 used scipy QUADPACK and failed the
    comparison with
# the recalled Newton formula at |x|/delta = 0.99 by 1.6e-7 (tolerance 1e-7), with QUADPACK
    reporting
# non-convergence for exactly that inner integral (integrable singularity at c = -1 when s ~ |x
    |).
# Run 2 changes ONLY the quadrature method (mpmath tanh-sinh, outer integral split at s = |x|).
# Tolerances, the formula and the bound are unchanged.
#
#   f_delta(x) = int_{|y-x|<delta} d^3y / (|y| |x-y|)
# Claim derived in the answer (AM-GM plus the bathtub comparison): f_delta(x) <= 4 pi delta for
     every x.
# Recalled comparison formula [R] (Newton's theorem): f = 4 pi (delta - |x|/2) for |x| <= delta
    ,
#   2 pi delta^2/|x| for |x| >= delta.
# Computed by direct quadrature centred at x: f = 2 pi int_0^delta s ds int_{-1}^{1} dc (|x|^2+
    s^2+2|x|s c)^(-1/2).
# Declared failure: any value exceeding 4 pi delta (1 + 1e-8), or disagreement with the
    recalled
# formula by more than 1e-7 relative.
import mpmath as mp
mp.mp.dps = 30
fails = []
print(" delta   |x|/delta   f (quadrature)      4 pi delta        Newton formula [R]   rel.diff
    ")
for delta in (mp.mpf('0.1'), mp.mpf(1)):
    for q in ('0', '0.1', '0.5', '0.99', '1', '1.5', '3'):
        X = mp.mpf(q) * delta
        def inner(s_):
            if X == 0:
                return mp.mpf(2)
            return s_ * mp.quad(lambda c: 1 / mp.sqrt(X**2 + s_**2 + 2 * X * s_ * c), [-1, 1])
        brk = [0, X, delta] if 0 < X < delta else [0, delta]
        f = 2 * mp.pi * mp.quad(inner, brk)
        newton = 4 * mp.pi * (delta - X / 2) if X <= delta else 2 * mp.pi * delta**2 / X
        rel = abs(f - newton) / newton
        print(f" {mp.nstr(delta,3):<6} {q:<10}  {mp.nstr(f,15):<18} {mp.nstr(4*mp.pi*delta,15)
    :<17} {mp.nstr(newton,15):<20} {mp.nstr(rel,2)}")
        if f > 4 * mp.pi * delta * (1 + mp.mpf('1e-8')):
            fails.append(("bound", float(delta), q))
        if rel > mp.mpf('1e-7'):
            fails.append(("newton", float(delta), q))
print("FAILURES:", fails if fails else "none")
```

```
 delta   |x|/delta   f (quadrature)      4 pi delta        Newton formula [R]   rel.diff
 0.1    0           1.25663706143592   1.25663706143592  1.25663706143592     0.0
 0.1    0.1         1.19380520836412   1.25663706143592  1.19380520836412     0.0
 0.1    0.5         0.942477796076938  1.25663706143592  0.942477796076938    0.0
 0.1    0.99        0.634601716025138  1.25663706143592  0.634601716025138    0.0
```

```
 0.1    1          0.628318530717959  1.25663706143592  0.628318530717959    0.0
 0.1    1.5        0.418879020478639  1.25663706143592  0.418879020478639    0.0
 0.1    3          0.20943951023932   1.25663706143592  0.20943951023932     0.0
 1.0    0          12.5663706143592   12.5663706143592  12.5663706143592     0.0
 1.0    0.1        11.9380520836412   12.5663706143592  11.9380520836412     0.0
 1.0    0.5        9.42477796076938   12.5663706143592  9.42477796076938     0.0
 1.0    0.99       6.34601716025138   12.5663706143592  6.34601716025138     0.0
 1.0    1          6.28318530717959   12.5663706143592  6.28318530717959     0.0
 1.0    1.5        4.18879020478639   12.5663706143592  4.18879020478639     0.0
 1.0    3          2.0943951023932    12.5663706143592  2.0943951023932      0.0
FAILURES: none
```

5. Result

(a) For every $N \geq 1$ and $\beta > 0$, with $\Delta\tau = \beta/N$ and $\mathbf{r}_{N+1} \equiv \mathbf{r}_1$, the primitive trace is the bead integral

$$Z_N(\beta) = (2\pi\Delta\tau)^{-3N/2} \int_{\mathbb{R}^{3N}} d^3r_1 \cdots d^3r_N \, \exp\Big\{ -\sum_{j=1}^{N} \Big[\frac{|\mathbf{r}_{j+1} - \mathbf{r}_j|^2}{2\Delta\tau} - \frac{\Delta\tau}{r_j} \Big] \Big\}.$$

This is (A3), taken as the definition [A1]; the same formula in the notes' units is (A4).

(b) $Z_N(\beta) = +\infty$ for every $N \geq 1$ and every $\Delta\tau > 0$. Two regions of configuration space are responsible.

- Configurations in which any single bead approaches the origin. These carry infinite weight on their own (B9). Inside any bounded region they are the only source (B11), and they have no counterpart in the continuum.
- Configurations in which the whole chain moves off to infinity (B13). This divergence is already present for $V = 0$ and is shared by the exact trace.

The kinetic factors cannot compensate. They are bounded below near any configuration with a bead at the origin, and maximal on the fully collapsed configuration. With an inner cutoff η on every bead and the beads confined to B_L, the integral grows like $(2\pi\Delta\tau)^{-3N/2}\big(4\pi\eta^4 e^{\Delta\tau/\eta}/\Delta\tau\big)^N$ as $\eta \to 0$: the N-th power of the one-bead divergence, with no suppression from the kinetic factors at leading order (B16), confirmed for $N = 2$ by C2. The divergence is not an artifact of the bead representation: the operator T^N, $N \geq 2$, is defined only on the zero vector, and T is unbounded (B.6).

(c) The exact kernel $K_\tau(\mathbf{r}, \mathbf{r}')$ is finite and jointly continuous on $\mathbb{R}^3 \times \mathbb{R}^3$ for every $\tau > 0$, because $-1/r$ is in the Kato class (C3) and Simon's kernel theorem applies. At the origin it is given by (C5); for instance $K_1(\mathbf{0}, \mathbf{0}) = 0.91957$, confirmed by two independent routes in C4. The exact trace $\operatorname{Tr} e^{-\beta H}$ is $+\infty$ for every $\beta > 0$, from the large-volume region only (C7); over a bounded region the exact diagonal kernel integrates to a finite number, whereas its primitive counterpart is infinite for every N.

The theorem that justifies time slicing is the Trotter product formula (C8) and (C9). Its imaginary-time clause requires both summands to be bounded below (in Kato's 1978 version, positive). $V = -1/r$ violates exactly that hypothesis and no other: the pieces are self-adjoint, and $K + V$ is self-adjoint on $D(K) = D(K) \cap D(V)$ by Kato's theorem. Consequently the real-time product formula holds for hydrogen and the imaginary-time one is not available.

Assessment of H1 to H6:

Hypothesis	Verdict	Where
H1	closed paths: correct. Ground-state dominance and $E_0 \approx -\ln Z/\beta$: wrong for hydrogen in $\mathbb{R}^3$, since $Z = \infty$	p. 4; Part D, (C7)
H2	wrong for $V = -1/r$: $e^{-\Delta\beta V}$ is unbounded and the per-slice expansion fails near $r = 0$ for every $\Delta\beta$. Incomplete even for matrices (no error accumulation)	p. 5; Part D, C.6
H3	wrong as written: βV for $\Delta\beta V$ (p. 6, fixed implicitly on p. 7); a one-dimensional prefactor $m/2$ and scalar coordinates for a three-dimensional problem; minor typos	p. 6, also p. 8; Part D, (A4)
H4	correct, but not sufficient on its own	p. 8
H5	correct, unproven; proof (B1) to (B3), growth law (B6); p. 9 item 1 reads "$\to 0$", probably shorthand	p. 8, p. 9
H6	correct conclusion, unproven; proof (B7) to (B9), sharpened in (B16). The large-volume divergence is not mentioned	p. 9

Domain of validity: hydrogen with an infinitely heavy point nucleus, in $\mathbb{R}^3$, atomic units, any $N \geq 1$, any $\Delta\tau > 0$. The statements about Z_N hold for the bead integral as defined by [A1].

7. Risk register

Item	Type	What happens if it is wrong
(A3) taken as the definition of Z_N	[A1], load-bearing	If an operator trace were intended: B.6 shows it does not exist, so the conclusion of (b) only strengthens
Trotter theorem statement, semigroup clause "A and B bounded below" (Reed–Simon I, Thm VIII.31)	[V] through secondary sources; Reed–Simon itself {ver: no}	If the clause reads "positive" or "non-negative", the identification of the violated hypothesis is unchanged, since V is neither. If the theorem number is wrong, the locator is wrong but the content stands

Item	Type	What happens if it is wrong
Kato's 1978 formula, positive self-adjoint A, B (Simon 2017, Thm 18.2)	[V], partial: excerpts confirmed the number, the title and the "positive self-adjoint" hypothesis; full statement not displayed	The remark on Kato's version would need rewording; the main identification rests on (C9)
H self-adjoint on $D(K)$ (Kato's theorem)	[V] (Simon 2017, §7, from excerpts)	Hypothesis (ii) in C.6 would be unverified; this is low risk, being a classical theorem
Continuous kernel for Kato-class V (Simon 1982, Thm B.7.1)	[S] {ver: no}; supported by the Broderix–Hundertmark–Leschke abstract {ver: tool}	Finiteness at general points would rest only on the BHL abstract; finiteness at the origin is independently established by C4
Coulomb data at the origin: $1/(\pi n^3)$, the Sommerfeld factor, E_n, completeness	[V] (Fäldt and Wilkin, eqs. (19), (21)); E_n and completeness [R]	Route (i) of C4 would be wrong; route (ii) does not use these data, and the two agree to 5×10^{-7}
$l = 0$ boundary condition $u(0) = 0$ in C4 route (ii)	[A]/[R]	Route (ii) would compute a different self-adjoint extension; the agreement with route (i) argues against this
Identity theorem for real-analytic functions	[R]	Remark B.6 would be unproven; (b) does not depend on it
Spectral theorem and functional calculus used in (C6)	[R]	Very low risk; the bound-state argument gives (C7) independently, through [R] spectral data
Hydrogen spectrum $E_n = -1/(2n^2)$ with multiplicity n^2	[R]; $E_{1,2,3}$ confirmed by C4	Used only as an illustration in C.4
Newton's theorem closed form	[R]	Used only as a comparison inside C5
Feynman–Kac and Khas'minskii remark in C.6	[R] {ver: no}	Qualitative and not used
Ground-state dominance for the kernel (H1 assessment)	[R]	Qualitative and not used
Reading of the handwritten notes (βV on p. 6; p. 9 item 1)	reading risk	An assessment of H3 or H5 could be misattributed; see Open questions
Prior-knowledge contamination from the project document	declared	The auditor should weigh the derivation against the declared result

8. Sources

Used

1. Encyclopedia of Mathematics, "Trotter product formula", the statement following eq. (a1): the unitary form holds when $A + B$ is essentially self-adjoint on $D(A) \cap D(B)$; the semigroup form holds for positive t when A and B are bounded from below. https://encyclopediaofmath.org/wiki/Trotter_product_formula (the redirect page shows oldid 51065). {ver: tool}. Retrieval note: page text returned by web search; both statements read.
2. "An Introduction to Lieb's Simplified approach to the Bose gas", arXiv:2308.00290, Theorem A1.17: "Given two operators A and B on a Hilbert space such that A, B and A + B are self-adjoint, and bounded from below, then for t ≥ 0, e^{−t(A+B)} = lim (e^{−tA/N} $e^{\{-tB/N\})}$N", attributed there to Teschl (2014), Theorem 5.12. {ver: tool}. Retrieval note: theorem text read in a search-result excerpt; Teschl not retrieved.
3. "Monotonicity of the Relative Entropy and the Two-sided Bogoliubov Inequality in von Neumann Algebras", arXiv:2501.04564, Theorem B.3.7, which quotes Reed–Simon I, Thm VIII.31, for the unitary product formula under "A+B essentially self-adjoint on dom(A) ∩ dom(B)". {ver: tool}. Retrieval note: search-result excerpt.
4. M. Reed and B. Simon, Methods of Modern Mathematical Physics I: Functional Analysis, revised ed. (1980), Theorem VIII.31. {ver: no}. Not retrieved; the semigroup clause is recalled and confirmed only through items 1 and 2.
5. B. Simon, "Tosio Kato's work on non-relativistic quantum mechanics", arXiv:1711.00528 (2017), §7, Theorem 7.3 (Kato–Rellich), and the sentence after eq. (7.15): "any V ∈ $L^2(\mathbb{R}^3)+L^\infty(\mathbb{R}^3)$ is $-\Delta$-bounded with relative bound zero". {ver: tool}. Retrieval note: the PDF fetched in this session was truncated after §5; §7 was read in search-result excerpts that reproduce the statement of Theorem 7.3 and that sentence. Also §18, Theorem 18.2 (Kato's product formula). {ver: tool}, partial: excerpts confirmed the number, the title, and that the result concerns positive self-adjoint A, B (self-adjoint contraction semigroups); the full statement was not displayed.
6. G. Fäldt and C. Wilkin, "Bound and unbound wave functions at short distances", arXiv:quant-ph/9710049, §IV: eq. (19), $[\psi(k,0)]^2 = 2\pi\eta/(e^{2\pi\eta} - 1)$ with $\eta = -me^2/\hbar^2 k$ for attraction; eq. (21), $[\psi_n(0)]^2 = 4\alpha_n^3$; normalisations from eqs. (3) and (4). {ver: tool}. Retrieval note: the arXiv PDF was fetched and the equations and normalisations read. The arXiv page lists no journal reference.
7. K. Broderix, D. Hundertmark and H. Leschke, "Continuity properties of Schrödinger semigroups with magnetic fields", Rev. Math. Phys. 12, 181–225 (2000), arXiv:math-ph/9808004: abstract (continuous integral kernel for Kato-class scalar potentials, extending Simon 1982) and §6, "Continuity of the integral kernel of the semigroup". {ver: tool}. Retrieval note: abstract, introduction and table of contents read; the theorem number in §6 was not reached, because the fetched text ended in §2.
8. B. Simon, "Schrödinger semigroups", Bull. Amer. Math. Soc. (N.S.) 7, 447–526 (1982), Theorem B.7.1. {ver: no}. Retrieval note: the Project Euclid PDF returned a bot-check page; the AMS issue page confirmed only the title and pages.
9. Z. Shen, "Absolute continuity of periodic Schrödinger operators with potentials in the Kato class", Illinois J. Math. 45, 873–893 (2001), eq. (1.1), the definition of K_d. {ver: tool}. Retrieval note: search-result excerpt reproduces the definition.
10. The team's notes, `Integrales_de_Camino_de_Feynman_261017_174852_2.pdf`, pp. 4–9, the object of Part D. Read in full from the attachment.

Consulted

- arXiv:1801.03361, which quotes Reed–Simon II, Theorem X.15 (self-adjointness of $-\Delta+v$ for $v \in L^2 + L^\infty$, $n \le 3$). {ver: tool}, search excerpt.
- arXiv:math-ph/9808011 ("Twist positivity"), which states the Lie–Trotter formula for self-adjoint, bounded-below H_0 and V with essentially self-adjoint sum, citing "Theorem A.5.1 of [2]". {ver: tool}, search excerpt.
- J. Chung and Y.-J. Kim, SIAM J. Math. Anal. 45, 728–731 (2013), on Lieb–Loss, Analysis, Theorem 9.7 (Newton's theorem), which is used only for the comparison formula in C5. {ver: tool}, abstract.
- Other papers quoting the Gamow/Sommerfeld factor $2\pi\eta/(e^{2\pi\eta} - 1)$ (e.g. arXiv:1710.04103 §2.2, arXiv:1406.6908 App. A). {ver: tool}, search excerpts; consistent with item 6.
- The project document `claude/start-here.md`, read because the project instructions require it. It did not affect the derivation and is declared in §1.

Not consulted: the GitHub repository named in prompt0 §5.

9. Open questions

1. Confirm in Reed–Simon I (revised edition, 1980) that Theorem VIII.31 contains the semi-group clause with "A and B bounded below". This needs library access, and a `{ver: human}` entry would settle it.
2. Confirm the statement of Simon 1982, Theorem B.7.1 (a jointly continuous kernel for $V_- \in K_\nu$, $V_+ \in K_\nu^{\mathrm{loc}}$), through AMS or library access.
3. The notes' authors should confirm my reading of the handwriting: that the βV in the second line of p. 6 is a slip of the pen, and what p. 9 item 1 is meant to say.
4. An auditor should check the real-analyticity step in B.6. A numerical illustration of its mechanism: for a positive ψ supported away from the origin, $e^{-\Delta\tau K}(w\psi)$ evaluated at the origin is strictly positive.
5. A third route to $K_\tau(\mathbf{0},\mathbf{0})$: a Feynman–Kac Monte Carlo over Brownian bridges. It would also test finiteness away from the origin, which C4 does not.
6. A Monte Carlo estimate of $Z_N^{(L)}(\eta)$ for $N = 3$ or 4, to test the N-th power law (B16) beyond $N = 2$.
7. A question about the problem's intent, not a proposal. Since the exact $\operatorname{Tr} e^{-\beta H}$ is itself infinite in $\mathbb{R}^3$, H1's use of Z needs a finite quantity. Which one does the larger project mean by Z?

Provenance

One row per tagged claim, built from the tags in the text. Generated from `sources/provenance.md`.

One row per tagged claim in `answers/answer-2.md`, built from the tags in the text rather than from its Sources section. This is the file that answers "did the model derive this or recall it".

Tag meanings are in `templates/CONVENTIONS.md` §1. A claim is load-bearing if the result in `final/final.md` §5 changes when the claim is wrong. Equation numbers are those of `answers/answer-2.md`, which `final/final.md` keeps.

Summary

Tag	Rows	Meaning
`[D]`	27	Derived in the document from premises stated in it
`[A]`	5	Assumption, definition or convention
`[S]`	1	Standard result invoked with a locator, not retrieved
`[V]`	7	Checked against a source retrieved this session
`[R]`	11	Recalled; no locator confirmed
`[N]`	6 checks (C1–C6) plus the failed C5 run 1	Numerical or symbolic computation
`[?]`	0	—

Load-bearing items that are not `[D]` are the three below. All three are in the Risk register.

- The definition [A1].
- The Trotter-formula statement (C8)–(C9), `[V]` through secondary sources only; Reed–Simon itself is `{ver: no}`.
- The kernel theorem in C.3, `[S] {ver: no}`, which the retrieved Broderix–Hundertmark–Leschke abstract supports.

Two `[R]` entries are not in the Risk register: the proton radius in §4.1 item 6, and "$R_{nl}(0) = 0$ for $l \geq 1$" in C4's coverage note. Neither is load-bearing. Together they are why the `no_unflagged_recall` gate fails; see `meta.yaml`.

Derived claims — `[D]`

Eq. / §	Claim	Source	Locator	Checked by	Load-bearing
(A1)	Trace in the position representation, with the identity inserted between factors (formal for this V)	this document	—	—	yes
(A2)	One-slice matrix element $= \rho_0(\mathbf{r}_{j+1}, \mathbf{r}_j; \Delta\tau)\, e^{\Delta\tau/r_j}$; potential at the right-hand point	this document; free kernel from prompt0 §3a	—	C2 (its $N = 2$ integrand)	yes
(A3)	Primitive trace as a $3N$-dimensional bead integral with prefactor $(2\pi\Delta\tau)^{-3N/2}$	this document	—	none; the prefactor is a declared blind spot (§6)	yes
(A4)	The same formula in the notes' units	this document	—	—	no
Part A	Cyclic invariance of the bead labels; independence from the factor order in T	this document	—	—	no (used to single out bead 1)
(B1)	$e^x \geq x^3/6$ for $x \geq 0$	this document	—	C1 (lower-bound column)	yes
(B2)–(B3)	$I_a(\eta, \varepsilon) \geq (2\pi a^3/3)\ln(\varepsilon/\eta)$, so $\int_{B_\varepsilon} e^{a/r} d^3r = \infty$	this document	—	C1	yes

Eq. / §	Claim	Source	Locator	Checked by	Load-bearing
(B4)–(B6)	Growth law $I_a \simeq 4\pi\eta^4 e^{a/\eta}/a\,[1 + 4\eta/a + 20(\eta/a)^2 + \dots]$, with remainder bound	this document	—	C1	no (quantitative; the divergence rests on (B3))
(B7)–(B9)	Restricted-domain lower bound on the Gaussians, so $Z_N = +\infty$ for every $N \geq 1$ and $\Delta\tau > 0$	this document	—	none directly; C2 covers the $N = 2$ cut integral	yes
B.2	Localised forms: $Z_N^{(L)} = \infty$ for every L; $\rho_N(\mathbf{r}, \mathbf{r}') = \infty$ for $N \geq 2$; ρ_1 not locally integrable at $\mathbf{r}' = 0$	this document	—	—	yes (part (c) comparison)
(B11)	Inside a bounded region with a neighbourhood of the origin removed for every bead, the integral is finite	this document	—	—	yes (identifies the region)
(B12)	Composition law of free kernels, by completing the square	this document	—	C3 (symbolic and numerical)	yes (for (B13))
(B13)	$Z_N \geq \int d^3 r_1 (2\pi\beta)^{-3/2} = \infty$: the large-volume divergence	this document	—	C3 (input only)	yes
B.5	Kinetic weight is maximal when the beads coincide, and positive at $\mathbf{r}_j = 0$ for fixed neighbours	this document	—	—	yes
(B14)–(B16)	With a cutoff, $Z_N^{(L)}(\eta) \simeq (2\pi\Delta\tau)^{-3N/2}(4\pi\eta^4 e^{\Delta\tau/\eta}/\Delta\tau)^N$	this document	—	C2 ($N = 2$ only)	no (sharpening)
(B17)	Angular identity $16\pi^2 e^{-(r_1^2+r_2^2)/\Delta\tau} \sinh\kappa/\kappa$	this document	—	C2(a), Monte Carlo	no
(B18)	$1 - R \simeq (2\eta^2/\Delta\tau)(1 + 2\eta/\Delta\tau)$ at $N = 2$	this document	—	C2(b)	no
B.6	$D(T^N) = \{0\}$ for $N \geq 2$; T unbounded	this document (uses an `[R]` identity theorem)	—	none	no
(C1)–(C3)	$\sup_{\mathbf{x}} f_\delta(\mathbf{x}) \leq 4\pi\delta$ by AM–GM and the bathtub comparison, so $-1/r \in K_3$	this document	—	C5	yes (hypothesis of the C.3 theorem)
(C4)–(C5)	Explicit $K_\tau(\mathbf{0}, \mathbf{0})$ from bound-state and continuum data at the origin	this document from `[V]` and `[R]` inputs	—	C4 (two routes)	yes (finiteness at the origin)
C.3, below (C5)	The bound-state sum is at most $e^{\tau/2}\zeta(3)/\pi$ (step added for F1.1)	this document	—	C6	no (C4 route (ii) is independent)

Eq. / §	Claim	Source	Locator	Checked by	Load-bearing
C.3, below (C5)	The continuum integrand behaves like k at 0 and like $k^2 e^{-\tau k^2/2}/2\pi$ at ∞; free limit recovered	this document	—	C4 (control G1)	yes
(C6)	Jensen: $\langle \varphi, e^{-\beta H}\varphi\rangle \geq e^{-\beta\langle\varphi, H\varphi\rangle}$	this document (uses `[R]` functional calculus)	—	none	yes (for (C7))
(C7)	$\operatorname{Tr} e^{-\beta H} = +\infty$, from translated orthonormal test states	this document	—	none	yes
C.6 table	Hypotheses (i) and (ii) hold for K, V; (iii) fails for V	this document from C.1	—	—	yes
C.6	Real-time formula (C8) holds; approximants are unbounded, and for $N \geq 2$ defined only on the zero vector	this document	—	—	yes
Part D	Assessment of H1–H6	this document, from the notes	notes pp. 4–9	read against the PDF	yes (deliverable)

Assumptions — `[A]`

Where	Assumption	What fails if it is wrong	Load-bearing
[A1], Part A	(A3), the bead integral, is the definition of Z_N	If an operator trace was meant, it does not exist (B.6), and (b) only strengthens	yes
C2	Two independent, uniformly distributed directions have $\cos\theta_{12}$ uniform on $[-1, 1]$	(B17) would be wrong; the Monte Carlo in C2(a) does not rely on it	no
C3	$s, t > 0$	Nothing physical; imaginary-time steps are positive	no
C4	The $l = 0$ boundary condition $u(0) = 0$ corresponds to the three-dimensional H (also tagged `[R]`)	Route (ii) would compute another self-adjoint extension; agreement with route (i) argues against this	no
C4	$u'(0)$ from the one-sided second-order formula, valid for u smooth at 0	Lower-order convergence; the observed order 1.96–1.99 argues against this	no

Standard results — `[S]`

Where	Claim	Source	Locator	Verified	Load-bearing
C.3	e^{-tH} has a jointly continuous integral kernel for $V_- \in K_\nu$, $V_+ \in K_\nu^{\mathrm{loc}}$	Simon, Bull. AMS 7 (1982)	Theorem B.7.1	`{ver: no}` (Project Euclid returned a bot-check page)	yes (finiteness at general points)

Verified against a retrieved source — [V]

Where	Claim	Source	Locator	Verified	Load-bearing
C.1	$V \in L^2 + L^\infty(\mathbb{R}^3)$ is $-\Delta$-bounded with relative bound 0; Kato–Rellich gives self-adjointness on $D(K)$	Simon, arXiv:1711.00528 (2017)	§7, Theorem 7.3 and the text after eq. (7.15)	{ver: tool} (search-result excerpts)	yes ((ii) in C.6)
C.2	Definition of the Kato class K_d	Shen, Illinois J. Math. 45 (2001)	eq. (1.1)	{ver: tool} (search-result excerpt)	yes
C.3	Continuous kernel for Kato-class scalar potentials, extending Simon 1982	Broderix, Hundertmark, Leschke, Rev. Math. Phys. 12 (2000)	abstract; §6 title	{ver: tool}; theorem number not reached	yes (supports the [S] row)
C.3	$[\psi_n(0)]^2 = 4\alpha_n^3$, radial normalisation	Fäldt and Wilkin, arXiv:quant-ph/9710049	§IV, eq. (21); normalisation eq. (3)	{ver: tool}	no (C4 route (ii) is independent)
C.3	Sommerfeld factor $2\pi\eta/(e^{2\pi\eta} - 1)$, with $\eta = -me^2/\hbar^2 k$ for attraction	Fäldt and Wilkin, arXiv:quant-ph/9710049	§IV, eq. (19); normalisation eq. (4)	{ver: tool}	no (same)
C.6	Trotter formula: unitary form if $A + B$ is essentially self-adjoint on $D(A) \cap D(B)$; semigroup form if, in addition, A and B are bounded below	Encyclopedia of Mathematics, "Trotter product formula"; arXiv:2308.00290; arXiv:2501.04564	entry after eq. (a1); Thm A1.17; Thm B.3.7 (quoting Reed–Simon I, Thm VIII.31)	{ver: tool} (secondary sources); Reed–Simon itself {ver: no}	yes
C.6	Kato's 1978 product formula requires positive self-adjoint A, B	Simon, arXiv:1711.00528	§18, Theorem 18.2	{ver: tool}, partial (number, title and hypothesis read in excerpts)	no

Recalled — [R]

Where	Claim	In Risk register	Load-bearing
B.6	Identity theorem for real-analytic functions	yes	no
C.3	$E_n = -1/(2n^2)$	yes	no (C4 check G4 reproduces $n = 1, 2, 3$)
C.3	Completeness of the bound states plus Coulomb waves	yes	no (route (i) only)
C.4	Spectral theorem and functional calculus, for (C6)	yes	yes, low risk
C.4	The bound-state sum $\sum_n n^2 e^{\beta/2n^2}$ diverges (illustration)	yes (hydrogen spectrum)	no

Where	Claim	In Risk register	Load-bearing
C.6	Feynman–Kac and Khas'minskii, qualitative remark (Simon 1982, `{ver: no}`)	yes	no
Part D, H1	Ground-state dominance for the diagonal kernel	yes	no
§4.1 item 6	Proton radius about $1.6 \times 10^{-5} a_0$	no	no
C4 coverage	$R_{nl}(0) = 0$ for $l \geq 1$, and likewise for continuum partial waves	no	no
C4	Limit-circle at $r = 0$; $u(0) = 0$ selects the regular solution	yes (as `[A]`/`[R]`)	no
C5	Newton's-theorem closed form for f_δ	yes	no (used only for comparison)

Computations — `[N]`

Check	Tests	Inputs from	Would not catch	Outcome
C1	(B2), (B6) by direct quadrature	the potential factor of the problem	the step from Z_N to one bead; the prefactor; kinetic factors	pass
C2	(B17) by Monte Carlo; (B16) and (B18) at $N = 2$	the $N = 2$ integrand of (A3)	$N \geq 3$; the large-volume region; the prefactor	pass
C3	(B12), symbolic and quadrature	the free kernel	anything involving V	pass
C4	(C5) against finite-difference diagonalisation; free control; box size; bound states	route (ii) uses only the radial Hamiltonian	points other than the origin; off-diagonal elements	pass
C5 run 1	(C3) bound; [R] Newton formula	definition of f_δ	—	FAIL against the Newton formula (QUADPACK non-convergence); bound passed
C5 run 2	same, with tanh-sinh quadrature	same	whether K_3 is the right hypothesis for the cited theorem	pass
C6	F1.1 bound, including the $\tau \to 0$ limit	the sum computed directly	the continuum term; recalled data	pass

Independence record

This is the file to read first when revisiting this subproblem. It is the solver's account of what it knew going in, and of where the derivation could have leaned on that knowledge.

`/log` assembled it from `answers/answer-2.md`. Sections 1 (Prior-knowledge declaration) and 6 (Convergence check) are reproduced verbatim; the notes at the end are by `/log`. Both sections are identical in answer-1 and answer-2, because they were on the round-1 frozen list.

- Solver: Claude Opus 5.5 (`claude-opus-5-5`, self-reported, configured model id); effort "extra" (user-reported, from `prompts/prompt0.md` §0).
- Auditor: "Gemini", effort high, both self-reported in the report's run header; no code execution, no web.

1. Prior-knowledge declaration

1. Do I recognize the problem? Yes. The failure of the primitive (Trotter) approximation for an attractive Coulomb potential is a known obstruction in path-integral Monte Carlo. I recall it from three sources, none retrieved in this session, all with unconfirmed locators: Ceperley's review of path-integral methods (Rev. Mod. Phys. 67, 279 (1995)); the discussion of the Coulomb path integral and "path collapse" in Kleinert's book on path integrals; and Kole and De Raedt, Phys. Rev. E 64, 016704 (2001). The project instructions required me to read the project document `claude/start-here.md` at the start of the session. It told me that this subproblem belongs to a larger problem on the hydrogen ground state from imaginary-time path integrals, that it corresponds to the first chunk of the Kole–De Raedt paper, and that the project has chosen the Kole–De Raedt effective action as its method. A memory note on the user's workflow says the same. I did not read the project's plan, its claim map or any repository file.

2. What do I believe the standard result is?

   - (i) Written as a bead integral, the primitive trace is $+\infty$ for every N and every $\Delta\tau > 0$. The potential weight $e^{\Delta\tau/r}$ is not integrable at $r = 0$ in three dimensions, and the Gaussian kinetic factors are bounded below near the singular configurations, so they cannot compensate.
   - (ii) The exact kernel $\langle \mathbf{r}|e^{-\tau H}|\mathbf{r}'\rangle$ is finite and continuous, including at the origin, because $-1/r$ is in the Kato class (Simon's 1982 review).
   - (iii) For hydrogen in infinite space, the exact trace $\operatorname{Tr} e^{-\beta H}$ is also infinite, for an unrelated reason: infinitely many bound states accumulate at zero energy, and there is a continuum.
   - (iv) The imaginary-time Trotter product formula requires both summands to be bounded below, and $-1/r$ is not.

3. What is the standard route, and where is it presented? Prove a lower bound on the bead integral restricted to a neighbourhood of the singular configuration. Simon 1982 covers kernels, and Reed and Simon volume I, §VIII.8, covers the Trotter formula.

4. What am I anchored on? Section 3c gives no expected result. I am anchored on item 2 and on the notes' own conclusions (H4 to H6 assert the divergence). Before deriving anything, I wrote down the checks I intended to run: a radial growth law, bead integrals with a cutoff, a collapse configuration, the exact kernel at the origin, and a bound for the Kato condition. So the shape of the verification was chosen with the expected answer in mind. Item 2(iii) is what made me look for a large-r divergence of Z_N, which the problem statement does not mention.

6. Convergence check

Does the derivation reproduce the declared result? Yes, items 2(i) to 2(iv) of the prior-knowledge declaration are all reproduced, and nothing in the derivation contradicts them. Several results go beyond what I declared:

- the sharp statement (B16) that the kinetic factors do not change the leading growth;
- Remark B.6, that T^N is defined only on the zero vector;
- the observation that the real-time product formula does hold;

- the explicit value of $K_\tau(\mathbf{0}, \mathbf{0})$.

Steps where someone who did not already know the target might not have chosen as I did:

1. Separating the divergence at the origin from the divergence at infinity. This was prompted by my recollection that the exact hydrogen trace diverges (declared). Someone without that recollection would still reach (B13) from $w \geq 1$, but might not look for it.
2. The region D_ε, with the other beads in B_1. It is motivated by the need for a lower bound on the Gaussians. The radius 1 is arbitrary; any bounded set works.
3. The power 3 in (B1). This is forced by the measure $r^2 dr$, as stated in the text.
4. The substitution $u = a/r$ and the repeated integration by parts. This is the standard treatment of a Laplace-type integral. I carried three terms so that the check could test beyond leading order.
5. The thin-shell lower bound (B15) and the parameter θ. This was chosen after (B6) showed that the integral concentrates at the cutoff.
6. The prediction (B18) and its first correction. Both were derived from the shell structure and stated in the script before it was run (observed 1.027, predicted about 1.025).
7. The translated test states in (C7). I chose these because I knew the divergence is a volume effect. Someone without that knowledge would more likely use the bound-state sum, which needs recalled spectral data.
8. The statement of the Trotter theorem. I recalled it and then checked it against secondary sources. Once the statement is fixed, identifying the failing hypothesis is mechanical.
9. The explicit value at the origin, (C5). This needs recalled Coulomb data. Route (ii) of C4 does not.
10. The real-analyticity argument in B.6. I chose this technique knowing that it would close.

Would the derivation still run if the result were slightly different?

- A different power of r. For $V = -1/r^\alpha$ with any $\alpha > 0$, the same proof runs with $e^x \geq x^n/n!$ and $n\alpha > 3$, so $Z_N = \infty$ for every $\alpha > 0$. In contrast, the exact kernel is finite only for $\alpha < 2$ (the Kato condition), and for $\alpha = 2$ with strong coupling the continuum fails as well. The derivation tracks this change at the step where n is chosen, and I would have noticed.
- A different dimension. In dimension d the condition is $n > d/\alpha$. In one dimension, the notes' setting, $\int dx\, e^{a/|x|}$ also diverges.
- A different growth law. Had the law been $\eta^3 e^{a/\eta}$, check C1 would have failed (see its contrast column).
- A different sign of the potential. For a repulsive potential, the proof of (B9) breaks at $w \geq 1$, which is visible.
- A different prefactor. A wrong prefactor in (a), for example the one-dimensional power $m/2$, would not be detected by anything in (b) or in the checks, since all the checks are independent of the prefactor. That is a blind spot. The prefactor rests only on the given kernel and the counting in (A2) and (A3).

Notes by /log

Context. The solver ran in a Claude session attached to the user's Vibe Physics project. Before starting, the project instructions required it to read `claude/start-here.md`, which names the subproblem's place in P003, the Kole–De Raedt paper, and the method chosen for P003. The solver declares this in §1 above. It did not open `PLAN.md`, `paper/`, or any file of this repository

during the loop. prompt0 §5 forbade the repository, and the repository was first opened at `/log` time, after the user closed the loop.

Ordering. Web retrieval for the Trotter, Kato-class and Coulomb-data citations happened before the derivation was written. The declaration in §1 is the solver's own account of its prior. No published solution of this subproblem was retrieved.

Anchoring steps. The solver self-reported ten (§6, list 1–10). The auditor reported pass 8 as "PASS. No circularity detected" and listed none of its own. That is not independent corroboration:

- the auditor read the declaration before forming a view;
- stage 0 (blind solve) was not run;
- the auditor executed no code.

Perturbation test. No perturbation test was run by anyone. The solver's §6 describes how the derivation would respond to $V = -1/r^{\alpha}$, to dimension d, to the sign of V, and to a different growth law, but that is analysis rather than a run. `meta.yaml` notes that the decisive test and the perturbation variant for K1 were to be sent to the auditor in `[FOCUS]` after answer-1 existed. The `[FOCUS]` block of `prompts/audit-prompt-1.md` was left empty, so they were not sent.

Audit history

One round. The solver produced `answers/answer-1.md`. The auditor returned a ten-pass report with verdict ACCEPT and one finding. The solver then produced `answers/answer-2.md`, from which this document is derived. The stopping rule of CONVENTIONS §6 was met at round 1, which returned no BLOCKER and no MAJOR.

The auditor was “Gemini” with effort “high”, both self-reported in the report’s run header. It declared code execution: no, and web: no. The audit prompt (`prompts/audit-prompt-1.md`) went out with its `[FOCUS]` block empty.

Finding F1.1, a NIT against Part C.3: the bound on the bound-state sum was asserted without the step that justifies factoring out $e^{\tau/2}$. Accepted. The term-by-term step was added, together with check C6, which tests the inequality and its sharpness as $\tau \to 0$. No finding was rebutted, so `rebuttals_upheld` is zero and carries no information here.

In its response the solver recorded four observations about the audit, contesting no finding (`audits/findings.yaml`, S1.1 to S1.4):

- Pass 7 reports the citations as verified although the auditor had no web access.
- Pass 3 did not recompute the steps the result rests on most: (B7)–(B9), (B12)–(B13), (B14)–(B16), Remark B.6, and (C6)–(C7).
- The corrector prompt asks both for a full rewrite and for the frozen sections to be left alone. The solver treated the frozen list as controlling.
- With one NIT open, CONVENTIONS §5 allows either ACCEPT or ACCEPT-WITH-FIXES.

`answers/answer-2.md` was not itself audited. Its new material, the F1.1 step and check C6, has never been read by an auditor. That is recorded here rather than resolved.

Code

Every script as executed in rounds 1 and 2. The scripts and their verbatim outputs are quoted in §4.2 above. The files are in `assets/code/`, each with a `.out` file holding its output and a

`.coverage.md` file stating what the check would and would not have caught. Check C6, added in round 2 for finding F1.1, is `c6_f11_bound.py`; it is listed here because §10 of answer-2 is not reproduced in this document.

`c1_radial_growth.py`

Reproduced in §4.2.

`c2_collapse_N2.py`

Reproduced in §4.2.

`c3_chapman_kolmogorov.py`

Reproduced in §4.2.

`c4_exact_kernel_origin.py`

Reproduced in §4.2.

`c5_kato_class_run1_scipy.py`

Reproduced in §4.2.

`c5_kato_class.py`

Reproduced in §4.2.

`c6_f11_bound.py`

```
# C6 (round 2, for finding F1.1): the bound on the bound-state sum in (C5).
# Environment: Python 3.13.16, mpmath 1.3.0. Deterministic.
#
# Claim (answer section C.3, as amended for F1.1):
#   S(tau) = sum_{n>=1} exp(tau/(2 n^2)) / (pi n^3)  <=  exp(tau/2) * zeta(3) / pi    for every
     tau > 0,
# because 1/n^2 <= 1 for n >= 1 and exp is increasing, so exp(tau/(2n^2)) <= exp(tau/2) term by
     term.
# Also: as tau -> 0 both sides tend to zeta(3)/pi, so the ratio S/bound -> 1 (the bound is
    sharp there),
# and the n = 1 term alone already exceeds a fraction 1/zeta(3) ~ 0.83 of the bound.
# Declared failure (before running): any tau with S > bound (relative excess > 1e-25 at 30
    digits),
# or a ratio S/bound that does not approach 1 as tau -> 0 (|ratio - 1| > 1e-5 at tau = 1e-6).
import mpmath as mp
mp.mp.dps = 30
fails = []
print(" tau        S(tau)                        e^{tau/2} zeta(3)/pi          S/bound")
for tau in ('1e-6', '0.01', '0.5', '1', '2', '4', '10', '40'):
    t = mp.mpf(tau)
    S = mp.nsum(lambda n: mp.exp(t / (2 * n**2)) / (mp.pi * n**3), [1, mp.inf])
    B = mp.exp(t / 2) * mp.zeta(3) / mp.pi
    r = S / B
    print(f" {tau:<10} {mp.nstr(S, 20):<28} {mp.nstr(B, 20):<28} {mp.nstr(r, 12)}")
    if S > B * (1 + mp.mpf('1e-25')):
        fails.append(("bound", tau))
    if tau == '1e-6' and abs(r - 1) > mp.mpf('1e-5'):
        fails.append(("limit", tau))
print("FAILURES:", fails if fails else "none")
```

Output (`c6_f11_bound.out`):

```
 tau        S(tau)                        e^{tau/2} zeta(3)/pi          S/bound
 1e-6       0.38262676106338832345        0.38262678734451618617        0.999999931314
 0.01       0.38428093655714554674        0.38454451982513820772        0.999314557211
 0.5        0.47605805401776602442        0.49130227440473563068        0.968971809859
 1          0.59534417043728417131        0.63084460761217577588        0.943725543903
 2          0.94276943003957727785        1.0400869230766701787         0.906433307757
 4          2.4461191758354434537         2.827249383017193344          0.865193990501
 10         47.416373621049848177         56.78682187365138956          0.834989035424
 40         154432884.17555492033         185637107.23244564432         0.831907405141
FAILURES: none
```

References

[1] Trotter product formula. Encyclopedia of Mathematics. URL `https://encyclopediaofmath.org/wiki/Trotter_product_formula`. Locator: the statement following eq. (a1); unitary form for $A+B$ essentially self-adjoint on $D(A)\cap D(B)$, semigroup form for positive t when A and B are bounded from below. Verification: tool (page text from web search).

[2] An introduction to Lieb's simplified approach to the Bose gas. arXiv preprint, . Locator: Theorem A1.17 (attributed there to Teschl, 2014, Theorem 5.12). Verification: tool (theorem text read in a search-result excerpt). Author not recorded.

[3] Monotonicity of the relative entropy and the two-sided Bogoliubov inequality in von Neumann algebras. arXiv preprint, . Locator: Theorem B.3.7, which quotes Reed–Simon I, Theorem VIII.31, for the unitary product formula. Verification: tool (search-result excerpt). Author not recorded.

[4] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics I: Functional Analysis*. Academic Press, revised and enlarged edition, 1980. Locator: Theorem VIII.31. Verification: no (not retrieved; the semigroup clause is recalled and confirmed only through secondary sources).

[5] Barry Simon. Tosio Kato's work on non-relativistic quantum mechanics, 2017. Locators: §7, Theorem 7.3 and the text after eq. (7.15), verification tool (search-result excerpts; the fetched PDF was truncated after §5); §18, Theorem 18.2, verification tool, partial (number, title and the positive self-adjoint hypothesis read in excerpts).

[6] Göran Fäldt and Colin Wilkin. Bound and unbound wave functions at short distances, 1997. Locators: §IV, eq. (19) (Sommerfeld factor) and eq. (21) (bound-state value at the origin); normalisations eqs. (3) and (4). Verification: tool (arXiv PDF fetched). No journal reference is listed on the arXiv page.

[7] Kurt Broderix, Dirk Hundertmark, and Hajo Leschke. Continuity properties of Schrödinger semigroups with magnetic fields. *Rev. Math. Phys.*, 12:181–225, 2000. doi: 10.1142/S0129055X00000083. Locators: abstract; §6, "Continuity of the integral kernel of the semigroup". Verification: tool (abstract, introduction and contents read; the theorem number in §6 was not reached).

[8] Barry Simon. Schrödinger semigroups. *Bull. Amer. Math. Soc. (N.S.)*, 7(3):447–526, 1982. Locator: Theorem B.7.1. Verification: no (the Project Euclid PDF returned a bot-check page; the AMS issue page confirmed only the title and pages).

[9] Zhongwei Shen. Absolute continuity of periodic Schrödinger operators with potentials in the Kato class. *Illinois J. Math.*, 45(3):873–893, 2001. Locator: eq. (1.1), the definition of K_d. Verification: tool (search-result excerpt).

[10] The team. Integrales de camino de Feynman (handwritten notes, in Spanish), 2026. Locator: pp. 4–9 (the attempt at S01; pp. 1–3 are background). File: human/Integrales_de_Camino_de_Feynman_261017_174852_2.pdf. The object assessed in Part D.